Physics Analysis of an Arduino-Controlled Servo Catapult

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Grade 11 Physics (SPH3U1)

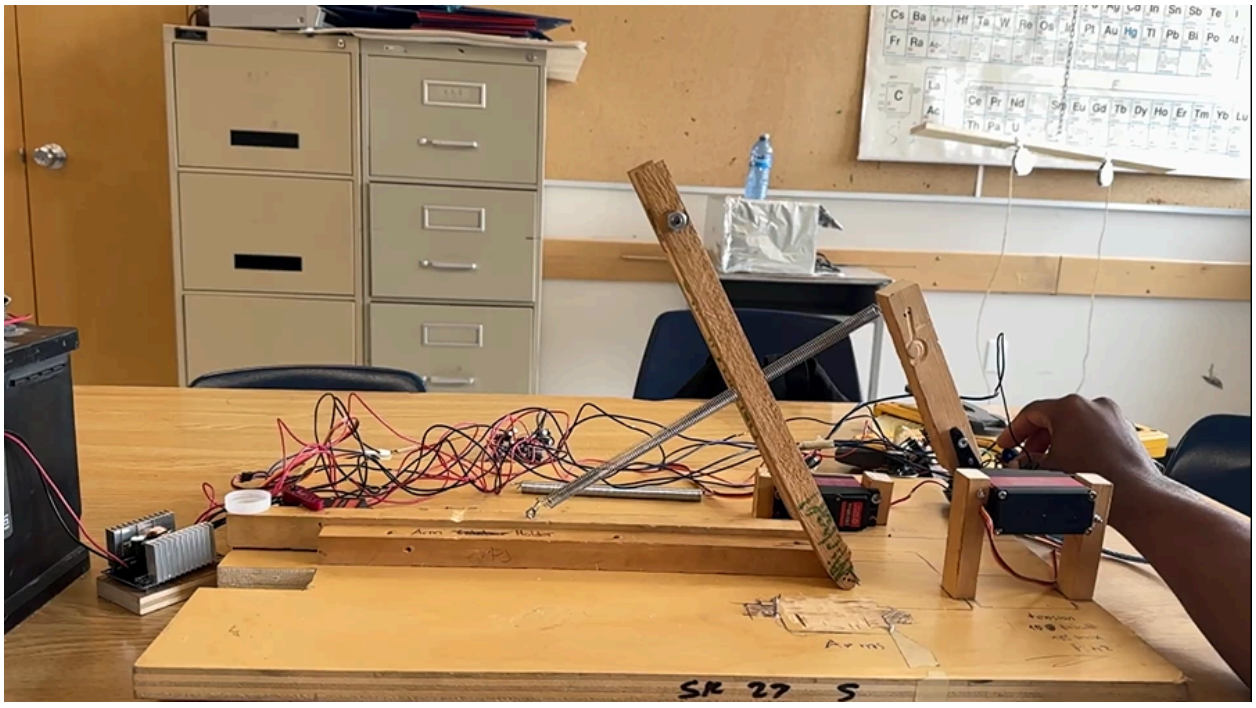
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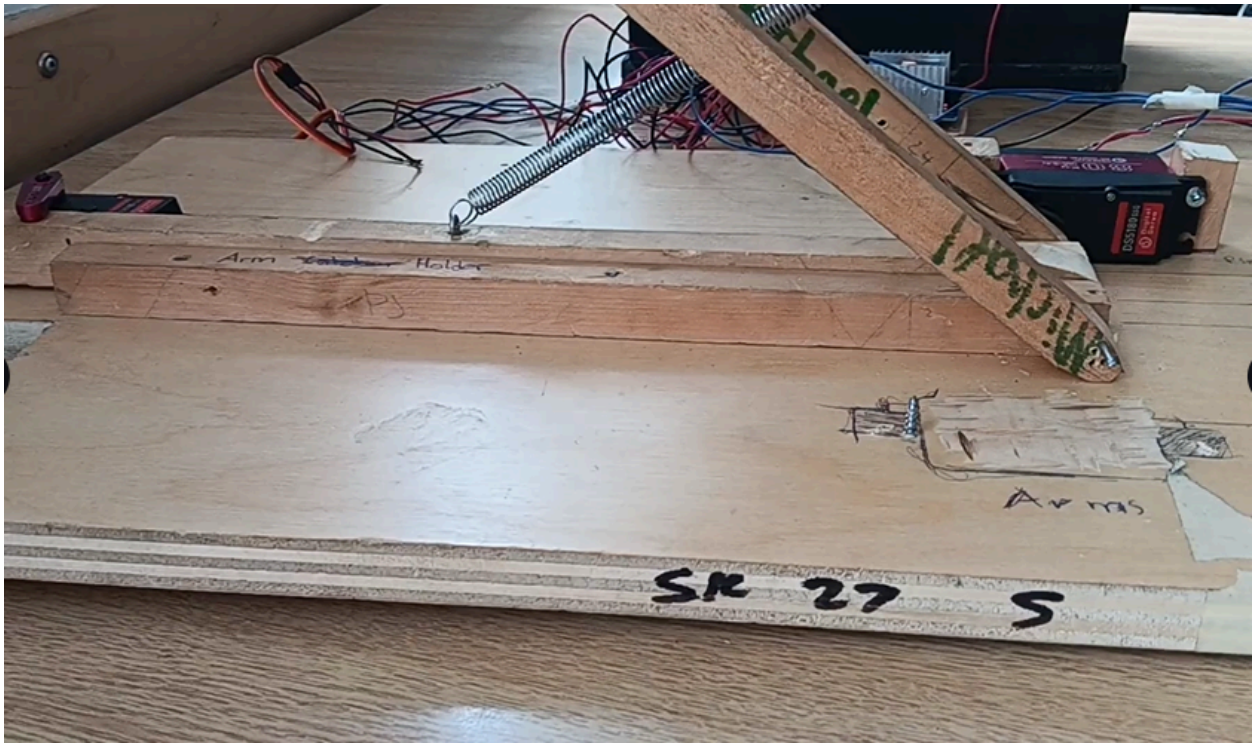
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This report examined an Arduino-controlled catapult by measuring its geometry, spring properties, electrical control settings, and projectile motion, then comparing two methods for predicting launch range across five trials. The spring constant was found to be $\boldsymbol{k_{avg}}$ **= 101.6609195** $\boldsymbol{N/m}$, and the five trial launch extensions ranged from **0.06** $\boldsymbol{m}$ to **0.238** $\boldsymbol{m}$, showing a clear increase in stored elastic energy as tension increased. A teacher-based trajectory method produced predicted ranges of **11.71** $\boldsymbol{m}$, **17.45** $\boldsymbol{m}$, **6.54** $\boldsymbol{m}$, **8.88** $\boldsymbol{m}$, and **8.35** $\boldsymbol{m}$, while the energy-based method produced **9.52** $\boldsymbol{m}$, **41.55** $\boldsymbol{m}$, **71.40** $\boldsymbol{m}$, **106.38** $\boldsymbol{m}$, and **79.23** $\boldsymbol{m}$ for Trials 1 through 5. The average percent difference between the two methods was **119.96254 %**, with the closest agreement in Trial 1 at **20.67407 %** and the largest difference in Trial 4 at **169.1812 %**. These results show that the catapult's motion can be modeled with projectile and energy physics, but real losses from arm rotation and other inefficiencies caused the two prediction methods to diverge significantly.

Completed project render:





(Double click to play video clips)

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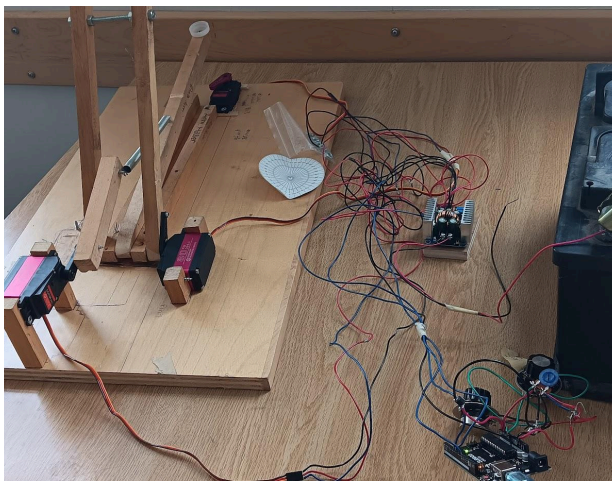
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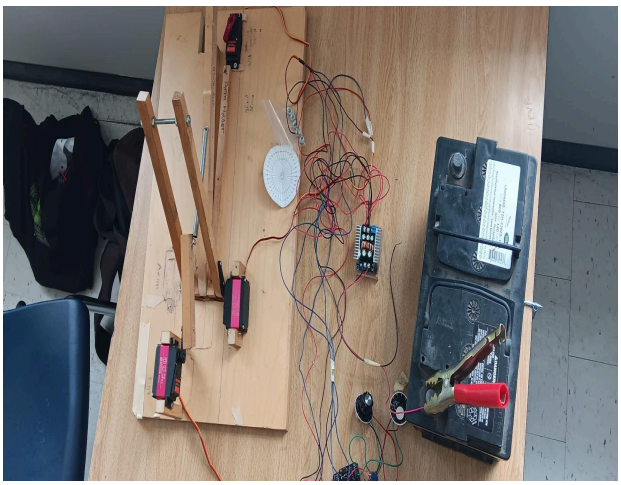
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Introduction

This project involves designing and building an Arduino-controlled catapult that launches small projectiles using a spring-powered arm. The system uses three servos: one to control the tension in the spring, one to set the stopping angle of the arm, and one to act as a trigger that releases the arm at the correct moment. Electronics were chosen for this project because of a love of programming and a desire to link it up with physics by constructing a catapult, which alone is filled with physics. This option is better, since it allows precise control over mechanical motion, which connects directly to key physics concepts that were learned in class such as energy conversion, forces, and motion. Electrical energy from the battery is converted into mechanical energy through the servos, while the Arduino acts as a control system that manages timing and positioning using inputs from potentiometers and switches. The main goals of this project are to analyze projectile motion, study how energy is transferred from the spring to the projectile, and understand the forces involved during launch. The project will involve building the system, testing different settings, calculating expected results using physics equations, and comparing those theoretical predictions with actual experimental data. Enjoy!!



(Fig 1: circuit path from side)



(Fig 2: circuit path from side)

Design and Construction

Before starting the building stage, it is important to familiarize with what is to be assembled. Starting with a sketch or a simple design; helps identify what is really needed, what electronics are doing what, and overall have a render of how everything will be finalized. Then gathering the materials needed helps stay organized, and have a visual of what else is needed, replaced, or discarded.

In this case, lucky enough to find a similar project on instructables:

Arduino Controlled Catapult : 7 Steps (with Pictures) - Instructables

With this blueprint ready, it is now possible to assemble everything and show a step by step process.

- Materials and Components List:

To build this automated catapult, the materials were divided into two main categories: the wooden structure and the electronic control system.

Wood and Mechanical Parts

- Base & Tower Wood: Solid wood boards used to build a strong, heavy platform and the upright support towers.
- Screws, bolt, and nuts: Hardware used to hold everything together and act as spin axes for the moving parts.
- Extension Springs: Strong metal springs used to store mechanical energy and pull the launch arm forward.
- Bottle Cap: A simple plastic bottle cap screwed to the end of the arm to act as the holder (launch pad) for the projectile.
- Metal Pivot Rod: A smooth metal axle that allows the main launch arm to swing back and forth easily.

Electronic components:

- Arduino Board (Elegoo UNO R3): The main controller responsible for running the code and controlling everything.
- Servos (Motors): Two large 80kg high-torque servos (one used to adjust the arm angle, and one used to pull the tension spring). One standard servo used to pull the launch trigger.
- Potentiometers (Control Knobs): Three twist knobs that let us manually adjust the arm angle, spring tension, and launch trigger.
- Reset Button: A 4-pin push button used to instantly reset all the motors back to their starting positions for the next launch.
- 12V DC-to-DC Buck Converter: A voltage regulator that drops the high battery power down to a safe level for the motors.
- 12V Car Battery: Provides plenty of electrical current to power the heavy-duty servos without draining the Arduino.
- Connecting Wires: Standard electronics wires soldered together to connect all the pieces to the Arduino.

- Arduino Code (written in C++):

```
#include <Servo.h>

Servo servo1; // Arm Angle Servo
Servo servo2; // Tension Servo
Servo servo3; // Trigger Servo

const int pot1Pin   = A1;
const int pot2Pin   = A2;
const int pot3Pin   = A3;
const int resetPin  = A4;

const int servo1Pin = 6;
const int servo2Pin = 5;
const int servo3Pin = 9;

// STATE MACHINE VARIABLE: false = Locked/Safe, true = Active/Unlocked
bool systemActive = false;
```

```
void setup() {
  servo1.attach(servo1Pin);
  servo2.attach(servo2Pin);
  servo3.attach(servo3Pin);

  pinMode(resetPin, INPUT_PULLUP);

  // Start perfectly safe and locked
  servo1.write(100);
  servo2.write(5);
  servo3.write(0);
}

void loop() {
  // 1. ALWAYS CHECK THE RESET BUTTON FIRST
  if (digitalRead(resetPin) == LOW) {
    systemActive = false; // Lock the system immediately!

    // Step-by-step physical relaxation sequence
    servo1.write(100);
    servo2.write(5);
    delay(1000);
    servo3.write(0);

    delay(200); // Debounce
  }

  // 2. SAFETY HANDSHAKE: If locked, check if Pot 3 is turned to 0 to
unlock
  if (systemActive == false) {
    int pot3Check = analogRead(pot3Pin);

    if (pot3Check < 50) { // If the trigger knob is turned all the way
down
      systemActive = true; // Unlock the system!
    }
  }

  // 3. EXECUTE CODES BASED ON SYSTEM STATE
```

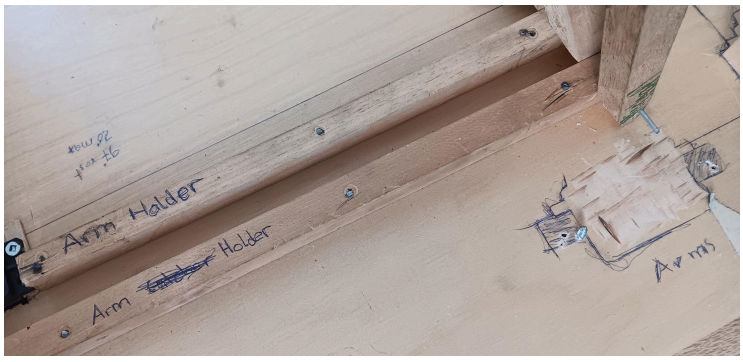
```
  if (systemActive == true) {
    // NORMAL OPERATION: Knobs are live
    int pot1Value = analogRead(pot1Pin);
    int pot2Value = analogRead(pot2Pin);
    int pot3Value = analogRead(pot3Pin);

    int angle1 = map(pot1Value, 0, 1023, 100, 20);
    int angle2 = map(pot2Value, 0, 1023, 5, 180);
    int angle3 = map(pot3Value, 0, 1023, 0, 180);

    servo1.write(angle1);
    servo2.write(angle2);
    servo3.write(angle3);
  } else {
    // LOCKED MODE: Keep forcing safe positions, ignoring Pot 1 and Pot 2
    servo1.write(100);
    servo2.write(5);
    servo3.write(0);
  }
  delay(15);
}
```

Step 1: Creating the base/framework

The wooden framework is the foundation of the whole machine. Cut and build it carefully so it can handle the heavy snap of the springs without breaking apart.



(Fig 3: Main arm support bars)



(Fig 4: tension servo mount)

1. Measuring and Cutting: Using a pen and tape measure, the dimensions were marked out on the wood. A bandsaw was used to safely cut the main baseboard and the vertical tower pieces to size.
2. Cutting on the Bandsaw: Take the marked wood to the bandsaw machine. Cut the long baseboard down to size, and then cut the four smaller pieces needed for the vertical catching arms.
3. Sanding: All the rough edges and cut lines were smoothed down using a wood sander machine to make sure the pieces fit perfectly flush together without splinters.
4. Assembling the Towers: Two vertical wooden bars were screwed securely into the main baseboard: the Arm Holder Tower and the Tension Tower. Screwing them tightly prevents the frame from bending or warping during the experiment.

Step 2: Mechanism building

The mechanical core of the catapult consists of a moving lever arm connected to an adjustable spring system.



(Fig 5: main pivot axle rod)

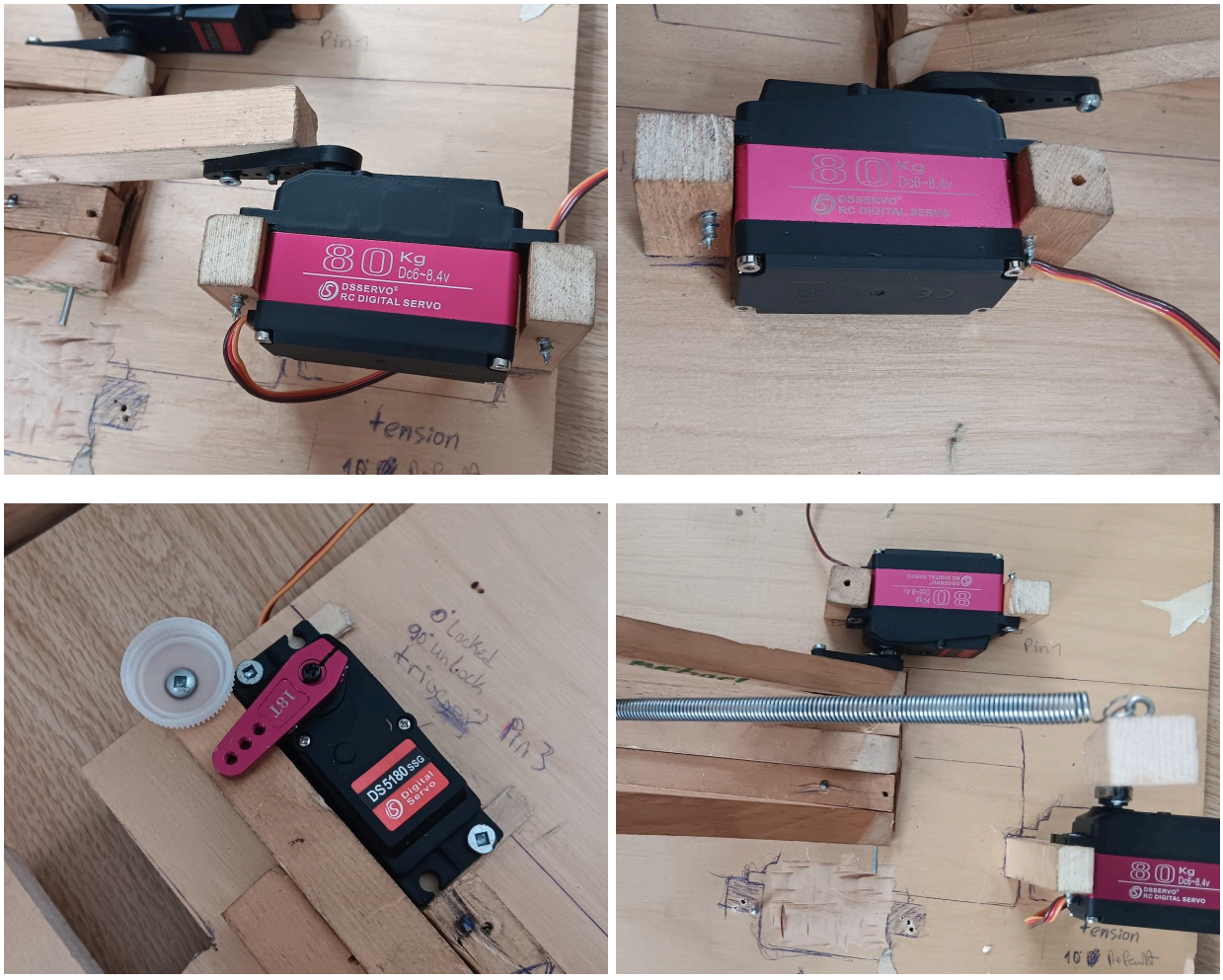

(Fig 6: Catching arms and main arm pivot)

1. Drilling the Pivot Hole: The main launch arm was marked where it needed to rotate. A power drill was used to make a perfectly straight hole through the arm and the support towers.

2. Installing the Axle: A smooth metal rod was cut down to size using a grinder to smooth out the sharp edges. This rod was pushed through the drilled holes to act as the main axle. It was secured using nuts tightened with pliers so the axle cannot slide out.

Step 3: Servo mounting positions

Because the two main servos are extremely powerful, they had to be mounted very securely to the wooden frame so they would remain tightened during the experiment..

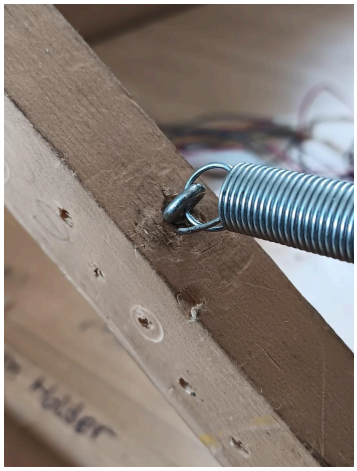


1. Arm Angle Servo (Servo 1): Screwed directly inside the front Arm Holder frame. This motor changes the angle of the catching arm, which alters the flight path of the projectile.

2. Tension Servo (Servo 2): Mounted flat on the back of the main wooden base. It is connected directly to the launcher's springs, winding them tight or letting them loose to control the launch power.
3. Trigger Servo (Servo 3): Mounted near the bottom of the frame. Since it only needs to pull a small lock out of the way to launch, this smaller micro-servo was mounted using a simple plastic bracket.

Step 4: Spring & cap attachment

This step connects the energy-storing springs and the projectile holder to the launch arm.

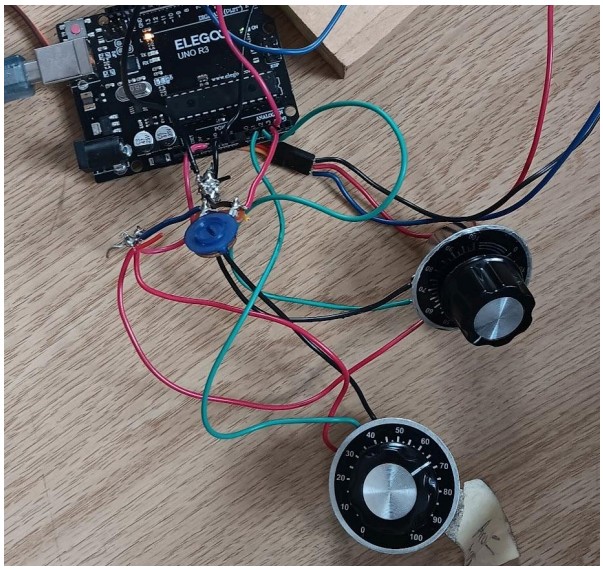




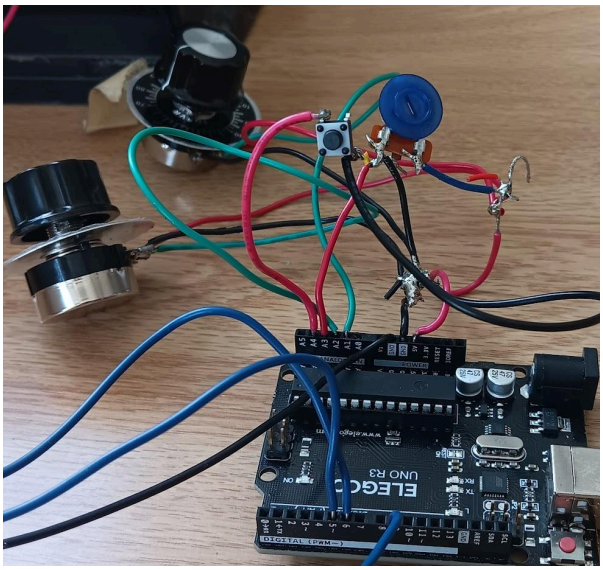
1. Attaching the Launch Cup: The plastic bottle cap was centered perfectly at the very top of the launch arm and screwed down. This creates a small, identical cup for the projectile to sit in every single time.
2. Hooking the Springs: Small metal eyelet screws were twisted into the middle of the wooden launch arm. The heavy steel springs were stretched and hooked directly into these eyelets using long-nose pliers.
3. Connecting to the Motor: The other end of the spring assembly was connected to the arm of the Tension Servo using a strong wire line, ensuring there is no loose slack when the motor is turned off.

Step 5: Arduino wiring and circuit Setup

The circuit was designed to safely route high-power electricity to the motors while keeping the control signals clean.



(Fig 7: Arms and tension potentiometer)

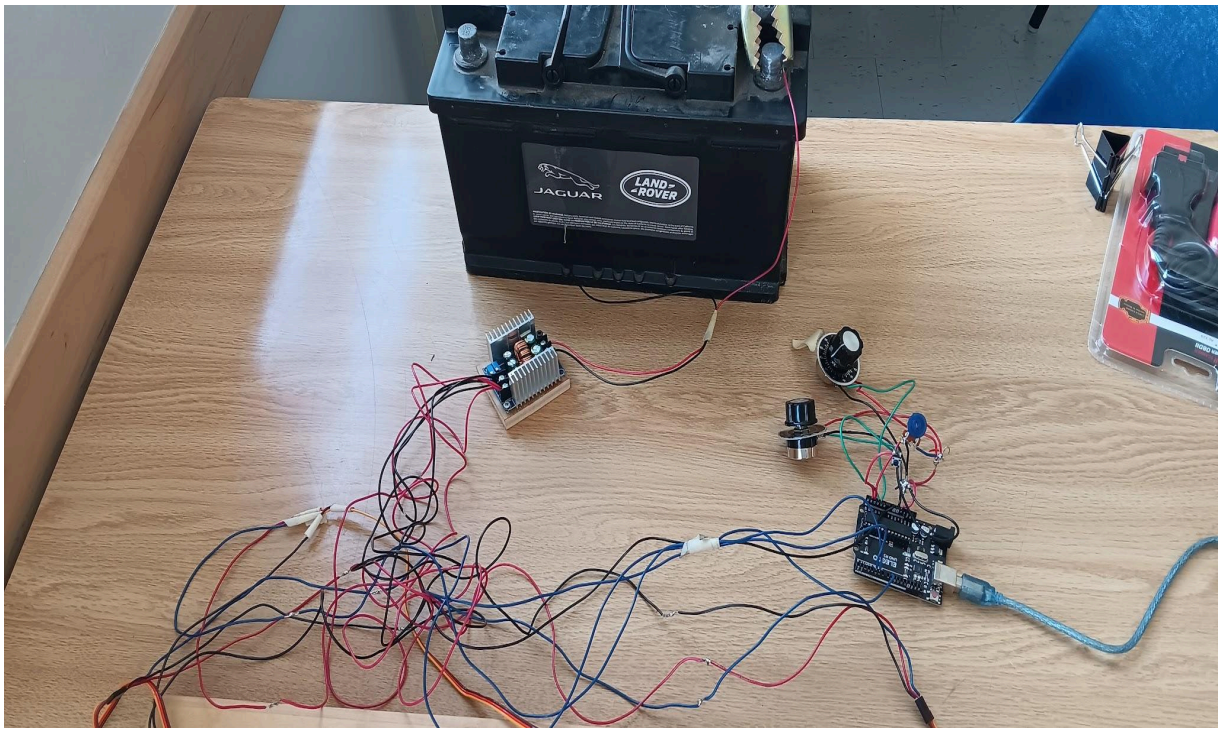


(Fig 8: trigger potentiometer, reset button)

1. Preparing the Wires: All the connection wires were measured, cut, and stripped of their plastic coating using wire strippers and side cutters.
2. Labeling the Knobs: The three potentiometers were wired up to share the same power and ground lines. To prevent confusion during testing, a piece of tape was stuck to each knob with its name written on it (like "angle arms") to know exactly which knob goes into Arduino’s pins A1, A2, and A3.
3. Soldering: A soldering iron and solder were used to permanently fuse all the wire connections together. During this process, a rusted pin on the trigger knob broke off, so it was ground clean and re-soldered to make a perfect connection.
4. Adding the Reset Button: The 4-pin push button was soldered diagonally across pin A4 and Ground, using the Arduino's internal programming to keep the circuit stable.

Step 6: Completed assembly and Logic

The final step combined the physical wooden project with the smart Arduino software.



(Fig 9: Complete circuit paths)

1. Dual Power Supply: The system uses two power sources to stay safe. A smartphone was connected into the Arduino via a USB/OTG cable to turn on the computer brain and pre-charge the capacitor cluster. Separately, a 12V car battery is connected to a buck converter to give the heavy motors their own isolated power supply.
2. The One-Click Reset Sequence: The code is programmed to protect the machine. When the Reset Button is clicked once, the Arduino instantly ignores the control knobs and forces the arm and tension servos to completely relax. The code pauses for exactly 1 second to give the heavy wooden parts time to physically drop back down, and only then does the trigger servo spin back to lock everything safely into place.

Component Overview

This section explains the role of each major part of the Arduino-controlled catapult and how they work together as one system. The catapult combines electrical control with mechanical motion: the Arduino reads inputs from potentiometers and a button, sends signals to servos, those servos move the mechanism, and the spring–arm system launches the projectile. Later sections of the report will use this overview as the basis for the physics analysis.

Arduino Board (Control Unit)

The Arduino is the main control unit of this project. It receives input from the potentiometers and switches, then sends output signals to the servos so the catapult can be tensioned, locked, and released in a controlled order. In simple terms, the Arduino acts like the "brain" of the catapult because it decides when each part should move.

For this project, the Arduino does not launch the projectile by itself. Instead, it controls the timing and position of the mechanical parts, which makes the launch more repeatable and easier to test. This is important because one of the goals of the project is to compare experimental launches with estimated or calculated values later on.

Servos

Servos are the parts that create the motion in the mechanism. A servo motor is useful because it can rotate to a specific angle when it receives a pulse-width control signal from the Arduino, which makes it ideal for a project that needs accurate movement . Each servo has a different job, so the system can be divided into separate actions instead of relying on one motor to do everything.

- Tension Servo

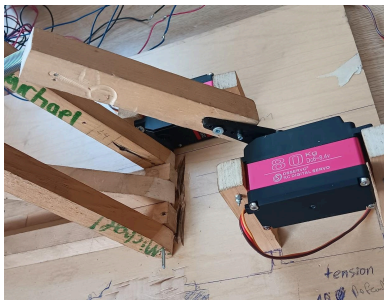
  The tension servo pulls on the arm attached to the spring, increasing the stretch in the spring and storing elastic potential energy. Its angle is controlled by a dedicated potentiometer, so turning that knob changes how far the spring is pulled back. This makes it possible to test different tension settings and see how they affect launch distance and projectile speed.

- Arm Servo (Stopping/Catching Servo)

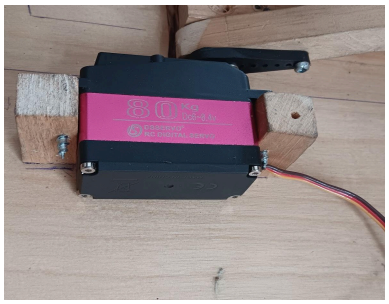
  The arm servo sets the stopping position for the catapult arm. It acts like an adjustable “catch” that limits how far the arm can rotate before it is released. Its angle is also controlled by a potentiometer, which lets us change the launch angle or stopping point without rebuilding the mechanism. This is important for studying how launch angle affects projectile motion.

- Trigger Servo

  The trigger servo is responsible for releasing the catapult arm at the right time. Its position is controlled by its own potentiometer. By turning this potentiometer, the Arduino rotates the trigger servo to move its axle that is holding the arm, allowing the arm to swing freely and launch the projectile. This design gives us fine control over how the trigger behaves and can be used to adjust the timing of the release.



(fig 10) - Tension servo



(fig 11) - Arms Servo



(fig 12) - Trigger Servo

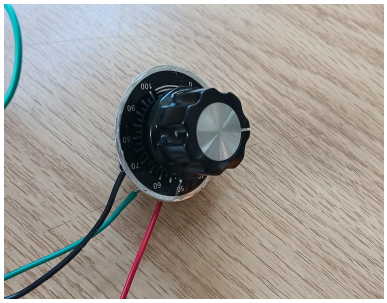
Together, these three servos turn the catapult into a programmable mechanical system. The Arduino can coordinate their positions so that tension is applied first, the arm is locked into place, and then the trigger is activated in a controlled sequence.

Potentiometers

Potentiometers are variable resistors that act as adjustable input controls for the Arduino. When connected as voltage dividers, they produce an output voltage between 0 V and 5 V depending on how far the knob is turned. The Arduino reads this voltage on an analog input pin and

converts it to a number, which it can then map to a servo angle. In this project, there are three potentiometers, each with its own function:

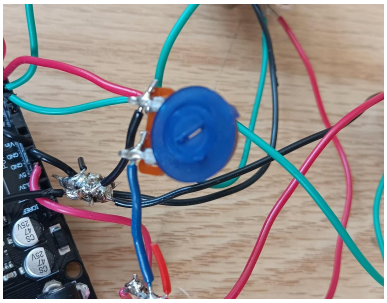
- One potentiometer controls the tension servo, setting how far the spring is pulled back.
- One potentiometer controls the arm servo, setting the catching angle or stopping position.
- One potentiometer controls the trigger servo, determining how and when the trigger moves to release the arm.



(fig 13) - Tension meter



(fig 14) - Arms meter



(fig 15) - Trigger meter

These three knobs serve as the user interface for the launch settings. Instead of hard-coding servo angles into the Arduino program, to be able to quickly adjust tension, angle, and trigger behavior by turning the potentiometers. This makes it easier to run multiple tests with different settings and compare how changes in the inputs affect the projectile's motion.

Push-Button (Reset Control)

The push-button is used as a reset control in the system. When the button is pressed, the Arduino moves all servos back to their default "ready" positions so the catapult can be prepared for the next launch. The button is wired as a digital input, and the Arduino checks its state (pressed or not pressed) in the main loop.

Using a button for reset has two advantages. First, it makes operation simpler: after a launch, pressing a single button to return everything to the starting configuration instead of manually turning each potentiometer back or moving the arm by hand. Second, it helps keep the testing process consistent, because the servos always return to the same default positions before the next trial.

Spring Assembly

The spring assembly is where mechanical energy is stored. When the tension servo pulls on the arm, it stretches the spring and increases its elastic potential energy. Once the trigger servo releases the arm, this stored energy is converted into rotational motion of the arm and then into kinetic energy of the projectile.

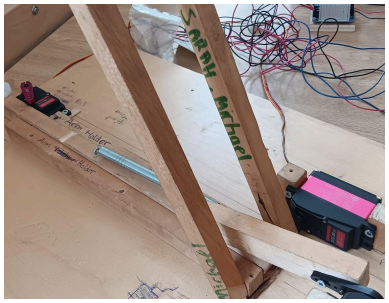
This part of the catapult provides the main link between the electronics and the physics of energy. The servos and Arduino determine how much the spring is stretched, while the spring itself is responsible for delivering the energy that actually launches the projectile. Later in the report, this assembly will be analyzed using ideas such as Hooke's law, spring constant, and elastic potential energy.

Catapult Arm and Frame

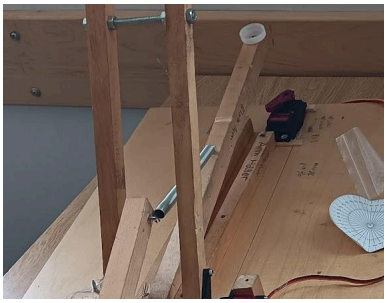
The catapult arm is the rotating lever that transfers energy from the spring to the projectile. It pivots around an axis fixed to the frame, secured by two side bars, to avoid the arm from spinning side by side when launching. When the arm is held back by the trigger and then released, it swings forward, pushing the projectile and giving it its initial velocity.



(fig 15) - Tension Arm



(fig 16) - Catching Arm



(fig 17) - Launch Arm

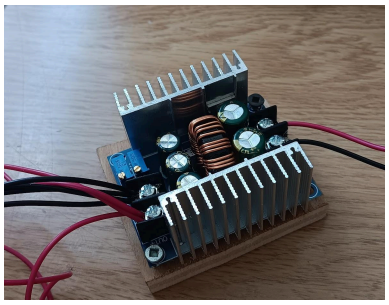
The frame supports all other components, including the servos, spring, and pivot. A large, strong, and stable frame is important because it resists the forces created during the launch and helps keep the geometry of the system constant from trial to trial. In the physics analysis, the arm can be treated as a rotating object with a moment of inertia, and its motion can be described using angular velocity, torque, and rotational kinetic energy.

Power Supply and Wiring

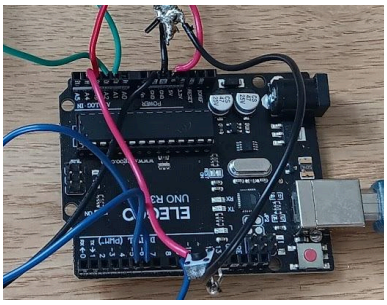
The power supply, in my case a 12V car battery, provides the electrical energy needed to run the Arduino and the servos. Servos can draw significant current, especially under load, so it is common to power them from an external source while still connecting their grounds to the Arduino ground so that the control signals work correctly. To safely do so, add a 12V DC-DC buck converter, that will control the power the servos draw from the battery. The wiring connects power, ground, and signal lines between the Arduino, servos, potentiometers, button, and power source.



(fig 18) - 12V Car Battery



(fig 19) - Buck Converter



(fig 20) - Arduino Board

Although the wiring itself does not move, it is crucial to the system. Without correct wiring, the control signals would not reach the servos, the potentiometers could not be read, and the button would not trigger a reset. In later sections of the report, this part of the system can be tied to basic electrical ideas such as voltage, current, and power.

System Interaction

When all components are combined, the catapult operates as a coordinated electromechanical system. Set the desired tension, arm angle, and trigger behavior using the three potentiometers. The Arduino reads these inputs, moves each servo to the appropriate position, and waits for the reset/launch button. The spring is stretched by the tension servo, the arm is held in place by the arm and trigger mechanisms, and then the trigger servo releases the arm so the projectile is launched.

This interaction between electronics and mechanics is what makes the project a “strong choice for a physics CPT“ -Jacobs. It allows the analysis of control signals, forces, energy transformations, and projectile motion in one setup, and creates many opportunities to compare theoretical predictions with experimental results.

Physics Theory

This project combines several important physics ideas: springs, energy, motion, forces, electricity, and rotation (ALL the topics we've covered in class). The catapult works by storing energy in a spring, holding that energy in place using the mechanical arm system, and then releasing it so the projectile can move forward. In this section, I will explain (the best I can, from what I've learned) the physics concepts behind the project before using them in the calculations section later on.

Energy in the Catapult

Energy is the ability to do work, and it can change from one form to another. In this catapult, electrical energy from the battery helps power the servos, mechanical energy is stored in the spring, and then that energy is transferred into motion when the arm is released. Some of the energy becomes the motion of the projectile, while some is lost as sound, heat, and friction.

The main idea behind the catapult is energy transfer. The tension servo pulls the arm and stretches the spring, which stores elastic potential energy. When the trigger releases the arm, the stored energy is converted into kinetic energy, which is the energy of motion. This is why the projectile can launch even though the Arduino itself is only controlling the servos and not producing the launch force directly.

In the calculations section, this energy relationship can be used to compare how much energy was stored in the spring and how much energy the projectile had after launch.

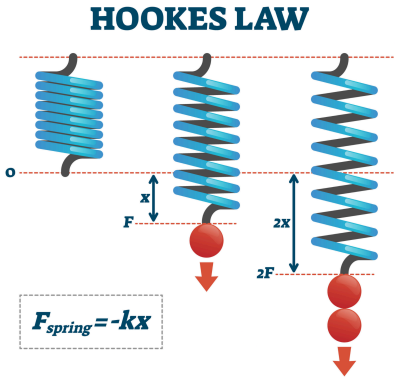
Springs and Hooke's Law

The spring is one of the most important parts of the catapult because it stores the launch energy. When a spring is stretched, it pushes back toward its original shape. This restoring force is described by Hooke's law, which says that the force applied to stretch a spring is proportional to the amount of stretch, as long as the spring is not stretched too far.

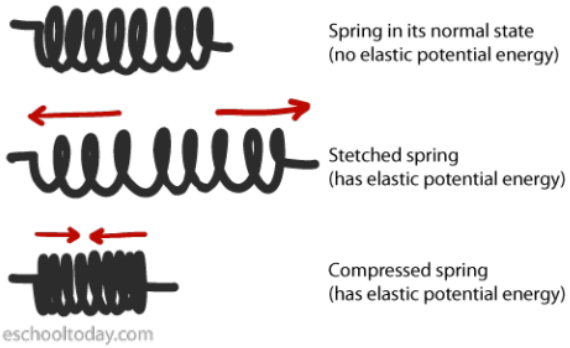
The formula for Hooke's law is:

$$F = kx \tag{1}$$

where F is the force in newtons, k is the spring constant, and x is the extension from the spring's natural length. A larger spring constant means a stiffer spring. That means it takes more force to stretch it the same distance. In the catapult, the tension servo stretches the spring by rotating the wooden arm connected to the servo axle, and the spring resists that stretch.



(fig 21) - Hooke's Law



(fig 22) - Spring stretched example

The energy stored in the spring is called elastic potential energy. It can be calculated using:

$$E_s = \frac{1}{2}kx^2 \qquad (2)$$

This formula is important because it tells us how much energy the spring can supply to the launch. The more the spring is stretched, the more energy it stores. In the calculations section, this equation will help estimate how much launch energy was available before the projectile was fired.

Rotational Motion of the Arm

The catapult arm does not move in a straight line. It rotates around a pivot, which means its motion is rotational instead of purely linear. Rotational motion is important because it helps explain how the spring's energy is transferred into motion before the projectile even leaves the catapult.

The rotational kinetic energy of the arm is:

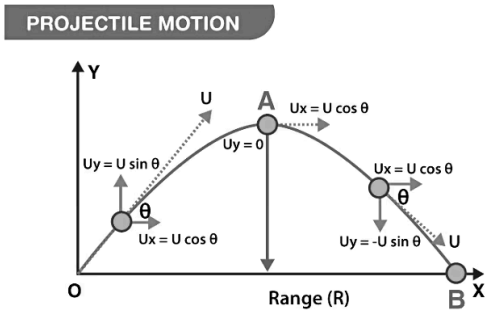
$$KE_{rot} = \frac{1}{2}Iw^2 \qquad (3)$$

where $\boldsymbol{I}$ is the moment of inertia and $\boldsymbol{w}$ is the angular speed. The moment of inertia describes how difficult it is to rotate an object, and it depends on how the mass is distributed around the pivot. If more mass is farther from the pivot, the moment of inertia is larger, and it takes more energy to rotate the arm.

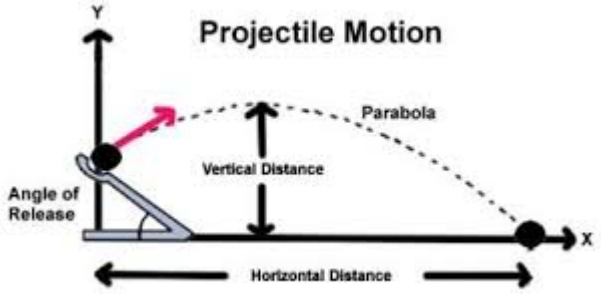
This matters in the catapult because not all of the spring's energy goes directly into the projectile. Some of it goes into rotating the arm itself. That is one reason why the projectile's actual speed may be lower than a perfect theoretical prediction. In the calculations section, rotational energy can be discussed as part of the energy lost or distributed through the system.

Projectile Motion

Once the load leaves the catapult arm, it becomes a projectile. From that point on, its motion is mostly controlled by gravity. The projectile moves forward horizontally while also moving up and down vertically, creating a curved path called a parabola.



(fig 23) - Projectile motion example



(fig 24) - Projectile launched

To study projectile motion, the launch velocity is split into two parts:

$$\boldsymbol{v_x = v_o \cos \theta} \quad (4)$$

$$\boldsymbol{v_y = v_o \sin \theta} \quad (5)$$

where $\boldsymbol{v_o}$ is the initial speed and $\boldsymbol{\theta}$ where is the launch angle. The horizontal and vertical parts of the motion can then be analyzed separately. The vertical motion is affected by gravitational acceleration, which is approximately $\mathbf{9.8 m/s^2}$.

The rotational kinetic energy of the arm is:

$$\boldsymbol{R} = \frac{{v_0}^2\, sin(20)}{g} \tag{6}$$

$$\boldsymbol{T} = \frac{2v_0\, sin\, \theta}{g} \tag{7}$$

These are used to predict the range and flight time of the projectile when it launches and lands at about the same height. In the calculations section, the measured range from the launch clips can be used to estimate the initial velocity of the projectile.

Forces and Motion

Forces are pushes or pulls that can change the motion of an object. In this project, forces are present when the spring stretches, when the arm rotates, and when the projectile is launched. Newton's second law states that:

$$\boldsymbol{F = ma} \tag{8}$$

where $\boldsymbol{F}$ is the net force, $\boldsymbol{m}$ is mass, and $\boldsymbol{a}$ is acceleration. This means that a larger force creates a larger acceleration if the mass stays the same.

Force is also important during launch because the projectile only gains speed while the arm is pushing it. Another useful idea is impulse, which connects force and time:

$$\boldsymbol{J = F\Delta t = \Delta p} \tag{9}$$

where $\boldsymbol{J}$ is impulse and $\boldsymbol{p}$ is momentum. This helps explain why a short but strong push from the catapult can launch a projectile quickly. In the calculations section, force and impulse can be used to estimate how much force acted on the projectile during the launch.

Electricity and Servo Control

The catapult is not only a mechanical system but also an electronic one. The Arduino sends control signals to the servos, and the servos use electrical power to move to specific angles. The three potentiometers and

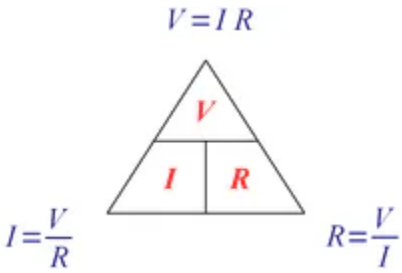
the push-button help control the positions of the servos and the reset process.

The basic electrical formulas that may be used later are:

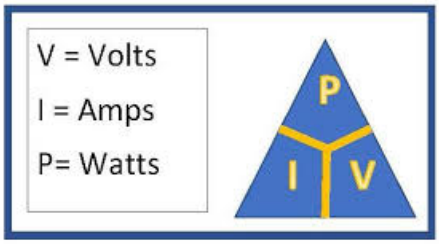
$$P = VI \tag{10}$$

$$E = Pt \tag{11}$$

where P is power, V is voltage, I is current, and t is time. These formulas can help estimate how much electrical energy is used by the system during a launch cycle. This is useful because it connects the electronics to the physics of motion and energy transfer.



(fig 25) - Ohms Law



(fig 26) - Watts Law

The servos are especially important because they turn electrical input into mechanical movement. Without them, the spring could not be tensioned or released in such a controlled way. That is why the electronics are a key part of the physics of the catapult, not just a separate piece of the project.

How This Theory Connects to the Calculations

The theory section explains the ideas, while the calculations section will use those ideas with real values from the project. For example, Hooke's law will be used with the spring's stretch to estimate force and elastic energy. Rotational energy will help explain how the arm moves and where some of the energy goes during launch.

Later, I will use my measured values for spring stretch, projectile mass, launch distance, and electrical input to compare theoretical predictions with actual launch results. This will show how well the physics equations model the real catapult system and where losses or experimental error may have affected the data.

Measurements and Data

This section collects all the raw measurements and basic processed data that will be used later in the calculations and analysis. It includes the physical dimensions of the catapult, masses of the projectiles and arm, basic electrical measurements, and selected settings of the potentiometers for tension and arm angle. Only simple operations such as unit conversions (for example, from centimetres to metres or from grams to kilograms) are done here; the actual physics calculations are left for the Calculations and Analysis section.

1. Catapult Physical Measurements and Projectiles

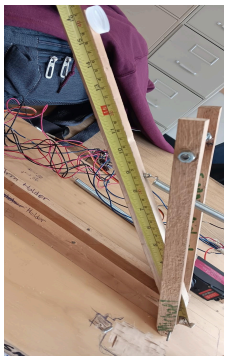
The goal of this subsection is to record the geometry and mass properties of the catapult and projectiles. These values will be used later to calculate energies, forces, and projectile motion.

All lengths were measured using a ruler or tape measure, and all masses were measured using a digital scale. Lengths were originally recorded in centimetres and then converted to metres where needed. Masses were measured in grams and converted to kilograms.

1.1 Tools used

- Ruler or tape measure (marked in centimeters)
- Digital mass scale (reads in grams)
- Pen and paper or a table in a document to record values

1.2 Measuring lengths (arm, pivot height, spring length)







1. Arm length (L_{arm}):

- Place the catapult on a flat surface.
- Identify the pivot point of the launch arm (the centre of the axle or rod the arm rotates around).
- Identify the projectile holder (where the nut rests on the arm - bottle cap).
- Align the zero mark of the ruler with the centre of the pivot.
- Measure the distance from the pivot to the centre of the projectile holder along the arm.
- Record this value in centimetres, then convert to metres by dividing by 100.

2. Pivot height (h_{pivot}):

- Place the catapult on the floor or a known reference surface.
- Hold the ruler vertically from the floor up to the pivot point.
- Measure the vertical distance from the floor to the centre of the pivot.
- Record this value in centimetres and then convert to metres.

3. Spring lengths (L_0 & L_{launch}):





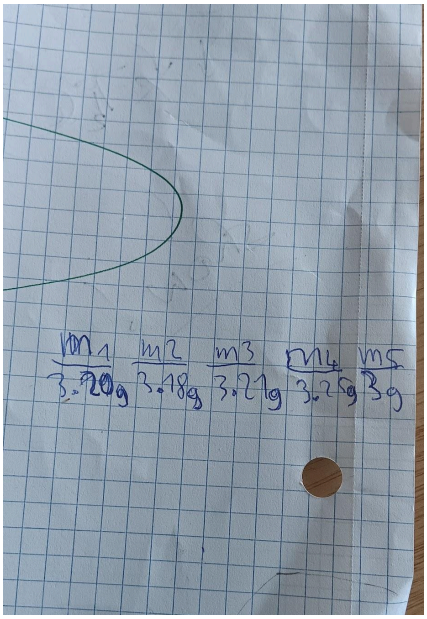
L_0: natural length of the spring with no tension
L_{launch}: length of the spring when the arm is pulled back to the launch position

- Ensure the catapult is not pulling on the spring (tension servo at minimum tension).
- Hold the ruler along the spring and measure the distance between the two eyelets where the spring is attached. Record this as L_0.
- Adjust the tension potentiometer so that the spring is stretched to the launch position you will use.
- Measure the distance between the same two eyelets again. Record this as L_{launch}

1.3 Measuring Mass







<u>Arm mass and projectile mass</u>

- Turn on the digital scale and zero it.
- Place the catapult arm alone on the scale (remove it if possible). Record the mass in grams and convert to kilograms by dividing by 1000.
- For each type of nut (projectile), place a single nut on the scale. Record its mass in grams and convert to kilograms.

1.4 (Table 1) - Physical Measurements of Catapult and Projectiles

Structural measurements				
Quantity	Symbol	Value (cm) / (g)	Value (m) / (kg)	Notes
Arm length (pivot to cap)	L_{arm}	40cm	0.4m	Measured along arm
Pivot height above floor	h_{pivot}	5cm	0.05m	Vertical from floor
Spring natural length	L_0	15cm	0.15m	No tension
Arm mass	m_{arm}	159.7g	0.1597kg	Digital scale
Projectile (nut) mass	m_{nut}	3.25g	0.00325kg	Digital scale

To save the data, record everything on an excel spreadsheet, which will later be used as a calculator to find values automatically, rather than manually writing multiple calculations.

2. Spring Extension Tests (Determining Average k)

This subsection shows how the spring constant k is found experimentally. The spring is tested vertically, off the catapult if possible, so that the only significant force stretching it is the weight of the hanging masses.

2.1 Tools used

- Spring from the catapult
- Clamp stand or fixed hook to hold the top of the spring
- Set of known masses (I used: 2.5kg, 2.0kg, 1.5kg)
- Ruler (cm)
- Spreadsheet software (Excel) for calculations

2.2 Setup and procedure (Vertical hanging test)





1. Remove the spring from the catapult if possible. Attach one end of the spring to a fixed support (for example, a hook or clamp stand) so that the spring hangs vertically.

2. With no mass attached, measure the spring length and record this as L_0 (in cm).

3. Hang a known mass m from the bottom of the spring. Wait until the spring stops bouncing.

4. Measure the new length L (in cm).

5. Repeat steps 3–4 for at least 2–3 different masses.

2.3 Calculating the Spring Constant k Using Hooke's Law

The spring constant k k describes how stiff the spring is. A larger k k means the spring needs more force to stretch by the same amount. In this project, k k is important because it will later be used to calculate the elastic potential energy stored in the spring before launch.

1. Manual calculation example for one trial

For one of the spring tests, the following values were measured:

$$L_o = 15 \text{ cm}$$

$$L = 44 \text{ cm}$$

First, the extension x of the spring is found by subtracting the natural length from the stretched length:

$$x = L - L_o$$

$$x = 44 \text{ cm} - 15 \text{ cm} = 29 \text{ cm}$$

Since Hooke's law uses SI units, the extension must be converted to metres:

$$x = 29 \text{ cm} = 0.29 \text{ m}$$

A mass of:

$$m = 2.5 \text{ kg}$$

was hung from the spring in this trial. The weight force of this mass is:

$$\vec{F}_g = m\vec{g}$$

$$F = (2.5\ \text{kg})(9.8\ \text{m/s}^2) = 24.5\ \text{N}$$

Hooke's law for a spring is:

$$F = kx$$

Substituting the expression for the weight force into Hooke's law gives:

$$mg = kx$$

Solving this equation for k:

$$k = \frac{(2.5\ kg)(9.8\ m/s^2)}{0.29\ m}$$

$$k = \frac{24.5\ N}{0.29\ m} \approx 84.5\ N/m$$

So, for this single trial, the spring constant is approximately 84.5 N/m 84.5 N/m. In practice, several trials with different masses are performed, and the final k k used in the report will be the average of the trial values.

2. Using an Excel spreadsheet to calculate k for multiple trials

Instead of repeating the same calculations by hand for each mass, a spreadsheet can be used to do the work automatically. This also makes it easier to include larger tables and graphs in the report.

A typical Excel layout for the spring test might be:

A (Mass kg)	B (g = 9.8)	C (Force N)	D (Lo m)	E (L m)	F (x m)	G (k N/m)
2.5	9.8		0.15	0.44		

1. Enter the measured values for each trial:

- Column A: mass m in kg
- Column D: natural length L_o in m
- Column E: stretched length L in m

2. In column C, use a formula to calculate the weight force:

- Example (row 2):
  =A2*B2
  This computes $F = mg$ in newtons.

3. In column F, calculate the extension:

- Example (row 2):
  =E2-D2
  This computes $x = L - Lo$ in meters.

4. In column G, calculate the spring constant for that trial:

- Example (row 2):
  =C2/F2
  This computes $k = F/x$ in N/m.

5. Copy these formulas down for all rows (all trials). Excel will calculate F, x, k automatically for each mass you tested.

6. At the bottom of column G, you can use the AVERAGE function to find the average spring constant:

- Example if trials are in rows 2–4:
  =AVERAGE(G2:G4)

This average value of k k will be the one used later in the Calculations and Analysis section to find the elastic potential energy in the spring and to estimate the launch speed of the projectile.

2.4 (Table 2) - Spring Extension Tests

SPRING TENSION AVERAGING

Trial	Mass m (kg)	Weight [m x Fg] (N)	Natural length Lo (m)	Stretched length L (m)	Extension x [L-Lo] (m)	Spring constant k [F/x] (N/m)
1	2.5	24.5	0.15	0.44	0.29	84.48275862
2	2	19.6	0.15	0.35	0.2	98
3	1.5	14.7	0.15	0.27	0.12	122.5
AVG						101.6609195

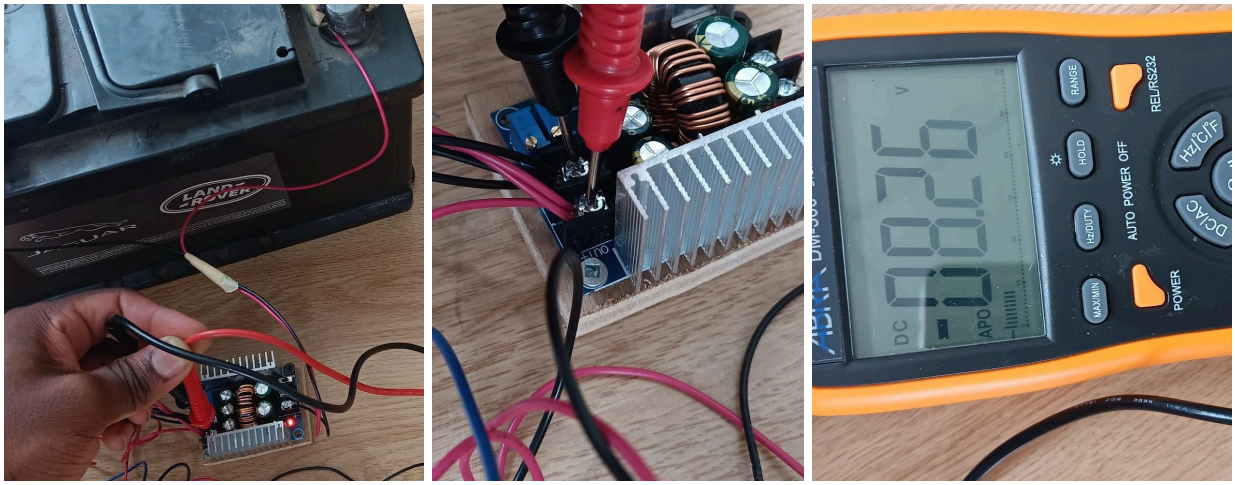
3. Electrical Measurements (Voltage, Current, and Timing)

This subsection explains how the basic electrical values were measured so that power and energy can be estimated later.

3.1 Tools used

- Multimeter (able to measure DC voltage and current)
- Access to the servo power wires (positive and ground)
- Stopwatch or video timer

3.2 Measuring servo supply voltage V



1. Turn on the catapult system so that the Arduino and servos are powered.

2. Set the multimeter to DC voltage mode.

3. Connect the multimeter's black (negative) probe to the servo power ground.

4. Connect the red (positive) probe to the servo power positive line (the same line that feeds the servos).

5. Read the displayed voltage value in volts (V) and record it in Table 3.

3.3 Measuring approximate servo current I

1. Set the multimeter to current mode (A or mA), with the correct range selected.

2. Disconnect the positive wire that supplies the servos.

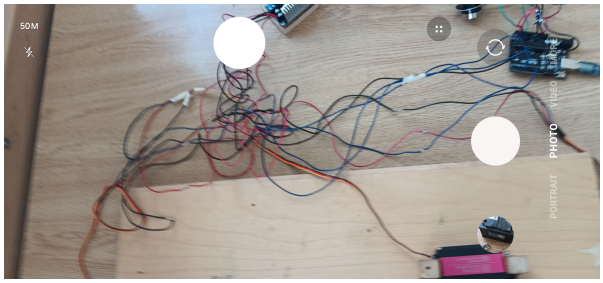
3. Insert the multimeter in series: connect one probe to the power source, and the other probe to the servo power wire, so that current flows through the meter.

4. Activate the tension or launch sequence so that the servos move.

5. Watch the multimeter display and note the approximate current while the servos are under load.

6. Record this value in amperes (A) or milliamperes (mA), converting to A as needed.

3.4 Measuring launch sequence duration t



1. Start a stopwatch or use a video recording with a visible timer.

2. Press the reset/launch button to begin the sequence.

3. Measure the time from the start of tensioning until the projectile is released.

4. Record this time in seconds (s) and note whether it includes only the tensioning phase or the full cycle.

3.5 (Table 3) - Electrical Measurements

Servo electrical power

Quantity	Symbol	Value	Unit	Notes
Servo supply voltage	V	8.26	V	Measured with multimeter
Approximate servo current	I	3.67	A	Series measurement during motion
Launch / tension duration	t	0.1658	s	Timing of launch sequence

4. Potentiometer Settings for Tension and Arm Angle

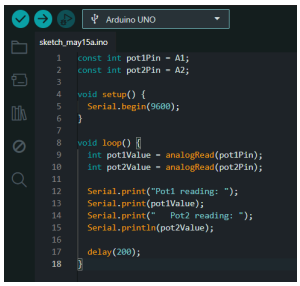
This subsection records the settings of the potentiometers that control the tension servo and arm servo for selected trials, so that later calculations can be matched to specific configurations.

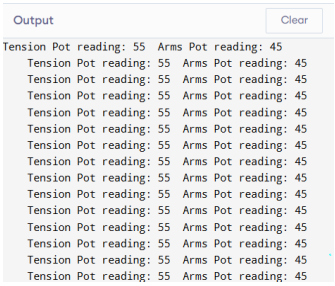
4.1 Tools used

- Arduino serial monitor (to read raw analog values)
- Protractor or video analysis to estimate arm angle

4.2 How tension potentiometer readings were recorded







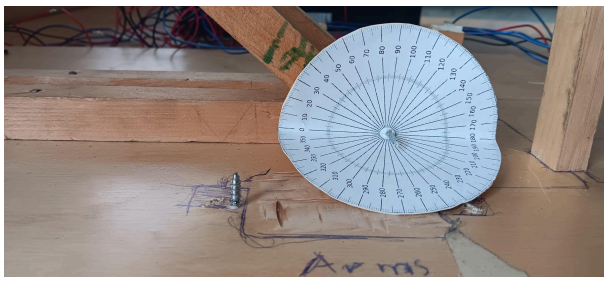
1. Connect the Arduino to a computer and open the serial monitor.

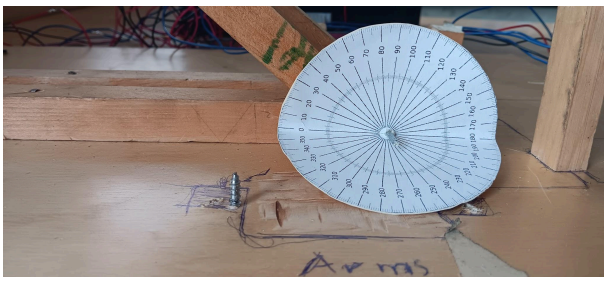
2. Modify the code (if needed) to print the analog reading from the tension potentiometer (for example, a value from 0 to 1023).

3. For each test tension setting, note the displayed analog value.

4. At the same time, measure the corresponding spring length L_{launch} as described in Section 1.2.

4.3 How arm angle θ was measured





1. Set the arm to its launch position for the trial.

2. Place a protractor at the pivot, with the baseline along the horizontal.

3. Measure the angle of the arm relative to horizontal when the projectile is released.

4. Alternatively, take a side-view photo or video and estimate the angle from the image.

5. Record the angle in degrees (°).

4.4 (Table 4) - Potentiometer Settings for Selected Trials

Potentiometers reading				
Trial	Tension pot reading	Spring length L (m)	Arm pot reading	Arm angle (*)
1	40	0.21	40	28
2	55	0.265	55	40
3	65	0.3	65	48
4	80	0.353	80	63
5	95	0.388	95	77

5. Launch Geometry for Ceiling Impact Trial

This subsection explains how the positions and heights in the ceiling-impact trial were measured, which are later used to determine the initial speed $|\boldsymbol{v_o}|$ and estimate the landing point.

5.1 Tools used

- Tape measure
- Ruler
- The classroom floor and ceiling as reference points
- Side-view video or photo (optional)

5.2 Measuring release point (x_1 , y_1)

1. Choose the launch point as the horizontal origin, so set $\boldsymbol{x_1}$ = 0.

2. Measure the vertical height of the projectile holder at the moment of release from the floor using a ruler.

3. Record this height as y_1 in centimetres and convert to metres.

5.3 Measuring ceiling impact point (x_2 , y_2)

1. After the projectile hits the ceiling, mark the impact point if possible.

2. Measure the horizontal distance along the floor (or height of the catapult's base if on an elevated surface) from the launch point to the point directly under the impact location. Record this as x_2, in centimetres, then convert to metres.

3. Measure the vertical distance from the floor to the ceiling at the impact location. Record this as y_2, in centimetres, then convert to metres.

5.4 Recording launch angle *θ*

1. Use the method in Section 4.3 to measure the arm angle at the moment of release.

2. Record the angle in degrees.

5.5 (Table 5) - Geometry of Ceiling Impact Trial

Test trial

Quantity	Symbol	Value	Unit	Notes
Release horizontal position	x1	0.27	m	Define as origin
Release vertical position	y1	0.32	m	Launch height above base
Ceiling impact horizontal position	x2	1.86	m	Distance from launch point
Ceiling impact vertical position	y2	2.13	m	Ceiling height at impact
Launch angle of arm	theta	35	*	Arm angle at release
Gravity magnitude	g	9.8	m/s^2	Standard gravitational acceleration

Calculations & Analysis/Discussion

This section brings together all the physics concepts from the earlier theory and the measurements collected from the catapult to answer one main question: where would the projectile actually land if it were launched in an open space? Instead of just doing random physics calculations, every step here is tied back to that question and to the real behaviour of the Arduino-controlled catapult.

The original plan for the experiment was to launch small projectiles (nuts) with different masses and different potentiometer settings for tension and arm angle, then measure where each nut landed and compare those results to theoretical predictions. During the first full test in the classroom, however, the projectile was launched with such a strong initial speed that it hit the ceiling before reaching the ground. That made it impossible to directly measure the true landing position inside the room. To solve this, the experiment was adapted: instead of measuring where the projectile actually landed, the landing point would be estimated using physics and then different methods would be compared.

The analysis therefore uses two main approaches. In Part A, a trajectory equation suggested in class by Mr. Jacobs is applied to the ceiling-impact trial. This equation relates the launch point, the ceiling impact point, the launch angle, gravity, and the initial speed $|\vec{v_0}|$. By rearranging this equation and substituting the measured values for the first trial, the initial speed of the projectile can be estimated, and from that, a theoretical landing distance $\boldsymbol{R}_{teacher}$ can be calculated. All the heavy numerical work (such as solving for $|\vec{v_0}|$ and calculating the range) is done using an Excel spreadsheet, with at least one full calculation shown by hand as an example.

In **Part B**, an independent estimate is created using the physics tools developed earlier in the report. First, Hooke's law and the measured spring constant k k are used to find the elastic potential energy stored in the spring before launch. Next, that energy is related to the kinetic energy of the projectile to estimate the initial speed $|\vec{v_0}|$ from an energy point of view. Projectile motion equations are then applied with this value of $|\vec{v_0}|$ and the measured launch angle θ to predict a landing distance $\boldsymbol{R}_{ours}$. Where possible, rotational motion of the arm and energy losses are

discussed to explain why the spring energy and projectile energy may not match perfectly. Again, Excel is used to perform repeated calculations quickly, while a few key examples are shown manually.

Although most of the earlier sections of the report focus on one main test trial (the ceiling-impact event), the analysis section extends this idea by looking at five different trial settings in Excel. Each trial corresponds to a different combination of tension potentiometer setting, arm angle, and sometimes projectile mass. For each trial, the "suggested" method (where it applies) and the "our method" predictions can be compared. This allows graphs to be created that show how the predicted landing distance changes with settings, and how closely the two methods agree across multiple cases. These graphs satisfy the requirement to include at least one graph in the report and help visualize the differences between the models.

Finally, **Part C** compares the results from both approaches and evaluates how consistent they are. The predicted landing distances from the suggested trajectory method and from the energy-based method are placed side by side, both for the main ceiling-impact trial and for the set of trials modeled in Excel. Any differences between $\boldsymbol{R}_{teacher}$ and $\boldsymbol{R}_{ours}$ are discussed in terms of the physics: energy lost to rotating the arm, friction at the pivot, air resistance, measurement uncertainties, and approximations made in the models. In this way, the Calculations and Analysis section does not just produce numbers, but also shows how the full chain of physics concepts: springs, forces, rotational motion, projectile motion, and electrical control; works together in the context of this specific catapult experiment.

Part A – Suggested Trajectory Method

This part uses Mr. Jacobs' trajectory equation to estimate the projectile's initial speed $|\vec{v_0}|$ from the ceiling-impact trial, then uses that speed to estimate where the projectile would have landed if the ceiling had not blocked it. This method is useful because it gives a direct estimate based on the actual launch path rather than on the spring alone.

A.1 The trajectory equation

For a projectile moving under gravity, the vertical position can be written as:

$$\boldsymbol{y_2} = \frac{g\Delta x^2}{2|\vec{v_0}|^2\, cos^2(\theta)} + \boldsymbol{tan(\theta)}\Delta x + \boldsymbol{y_1}$$

Where:

- $\boldsymbol{x_1}, \boldsymbol{y_1}$ is the launch point.
- $\boldsymbol{x_2}, \boldsymbol{y_2}$ is the point where the projectile hit the ceiling.
- $\Delta x = \boldsymbol{x_2} - \boldsymbol{x_1}$ is the horizontal distance traveled.
- $|\vec{v_0}|$ is the initial speed.
- $\boldsymbol{\theta}$ is the launch angle.
- $\boldsymbol{g}$ is the acceleration due to gravity.

This equation describes the path of a projectile in two dimensions. The first two terms show the upward or downward trend caused by the launch angle, while the last term shows the effect of gravity pulling the projectile downward.

A.2 Rearranging for $|\vec{v_0}|$

To find the initial speed, the equation must be rearranged so that $|\vec{v_0}|$ is isolated. Starting with:

$$\boldsymbol{y_2} = \frac{g\Delta x^2}{2|\vec{v_0}|^2\, cos^2(\theta)} + \boldsymbol{tan(\theta)}\Delta x + \boldsymbol{y_1}$$

First, move $\boldsymbol{y_1}$ term to the other side:

$$\boldsymbol{y_2} - \boldsymbol{y_1} = \frac{g\Delta x^2}{2|\vec{v_0}|^2\, cos^2(\theta)} + \boldsymbol{tan(\theta)}\Delta x$$

Then, move $\boldsymbol{tan(\theta)}\triangle x$ term to the other side:

$$\boldsymbol{y_2 - y_1 - + tan(\theta)}\triangle x = \frac{g\triangle x^2}{2|\vec{v_0}|^2\, cos^2(\theta)}$$

Then, move $\mathbf{2}cos^2(\theta)$ to the other side:

$$\boldsymbol{2cos^2(\theta)\cdot\ (\ y_2 - y_1 - + tan(\theta)}\triangle x\boldsymbol{)} = \frac{g\triangle x^2}{|\vec{v_0}|^2}$$

After, move $g\triangle x^2$ to the other side:

$$\frac{g\triangle x^2}{2cos^2(\theta)\ \cdot (\ y_2 - y_1 - + tan(\theta)\triangle x)} = |\vec{v_0}|^2$$

Now, solve for $|\vec{v_0}|$ by taking the square root:

$$|\vec{v_0}| = \sqrt{\frac{g\triangle x^2}{2cos^2(\theta)\ \cdot (\ y_2 - y_1 - + tan(\theta)\triangle x)}}$$

This is the equation that will be used in the spreadsheet for the teacher method.

A.3 Test Trial values

For the test trial, the known values are:

- $\boldsymbol{x_1}$ = 0.27 m
- $\boldsymbol{y_1}$ = 0.32 m
- $\boldsymbol{x_2}$ = 1.86 m
- $\boldsymbol{y_2}$ = 2.13 m
- $\boldsymbol{\theta}$ = 35°
- $\boldsymbol{g}$ = 9.8 m/s^2

The horizontal distance is:

$$\triangle x = \boldsymbol{x_2 - x_1 = 1.86\ m - 0.27\ m = 1.59\ m}$$

These values will be substituted into the rearranged equation to calculate $|\vec{v_0}|$.

A.4 Manual calculation example

The calculation can be shown in one worked example before being repeated in Excel. The setup is:

$$|\vec{v_0}| = \sqrt{\frac{(9.8)(1.59)^2}{2(cos^2\,35)\ \cdot (\,2.13 - 0.32 + (tan\,35)1.86)}}$$

Each part of the expression should be evaluated carefully in the correct order. Since the result depends on a few significant measured values, the spreadsheet will later be used to reduce arithmetic mistakes and make the process faster for multiple trials.

A.5 Excel setup for Part A

To speed up the calculations, Excel will be used to find $|\vec{v_0}|$ for the ceiling-impact trial and to repeat the process for the other trials. This avoids doing the same algebra and arithmetic by hand every time and makes it easier to compare values later.

1. Enter the measured values for each trial.

- Column A: Trial number
- Column B: $\boldsymbol{x_1}$ in m
- Column C: $\boldsymbol{y_1}$ in m
- Column D: $\boldsymbol{x_2}$ in m
- Column E: $\boldsymbol{y_2}$ in m
- Column F: $\boldsymbol{\theta}$ in degrees
- Column G: $\boldsymbol{g}$ in m/s^2

2. Calculate the horizontal distance.

In a new column, subtract the launch x-position from the impact x-position. Example for row 2:

=D2-B2 This gives:

$$\Delta x = \boldsymbol{x_2 - x_1}$$

3. Calculate the denominator inside the $|\vec{v}_0|$ formula.

You may want to create a helper column for:

$$y_2 \; - \; y_1 \; - \quad + \; tan(\theta)\Delta x$$

Example for row 2:

=C2 + H2*TAN(RADIANS(F2)) - E2 where $\boldsymbol{H2}$ is the Δx column.

4. Calculate $cos^2(\theta)$.

If needed, create another helper column. Example for row 2:

=COS(RADIANS(F2))^2

5. Calculate $|\vec{v}_0|^2$.

Use the rearranged formula in Excel. Example structure for row 2:

=(G2*H2^2) / (2*I2*J2) where:

- $\boldsymbol{G2}$ = gravity
- $\boldsymbol{H2}$ = Δx
- $\boldsymbol{I2}$ = $\cos^2(\theta)$
- $\boldsymbol{J2}$ = $y_2 \; - \; y_1 \; - \quad + \; tan(\theta)\Delta x$

6. Calculate $|\vec{v_0}|$.

Take the square root of the value from the previous step. Example for row 2:

=SQRT (K2)

7. Use $|\vec{v_0}|$ to calculate the predicted landing distance.

If you want the range formula in Excel:

$$\boldsymbol{R} = \frac{|\vec{v_0}|^2 sin(2\theta)}{g}$$

Example for row 2:

=(L2^2*SIN*RADIANS(2*F2))) /G2

8. Copy the formulas down for all remaining trials.

This allows Excel to calculate the same values for trials 2, 3, 4, and 5 automatically.

9. Find the final predicted teacher method range.

If you want to highlight the main trial, keep Trial 1 as the main reference point and use the other trials later for comparison if applicable.

10. (Table 6) - Suggested Excel table layout for Part A

Part A table

Heading	TRIAL 1	TRIAL 2	TRIAL 3	TRIAL 4	TRIAL 5
x1 (m)	0.339	0.252	0.246	0.148	0.088
y1 (m)	0.207	0.313	0.325	0.377	0.396
x2 (m)	3.238	2.2	3.261	8.021	8.021
y2 (m)	2.13	2.13	2.13	2.13	2.13
theta (*)	28	40	48	63	77
gravity - g (m/s^2)	9.8	9.8	9.8	9.8	9.8
delta x - dx (m)	2.899	1.948	3.015	7.873	7.933
y1 + dx tan(theta) - y2	-0.38157436	-0.18243392	1.543496732	13.69863251	32.62759811
cos^2(theta)	0.779596452	0.586824089	0.447735768	0.206107374	0.050602977
initial speed ^2 - Vo^2	-138.434225	-173.684261	64.45301063	107.5736335	186.7711609
initial speed - Vo (m/s)	11.76580747	13.17893248	8.028263239	10.37177099	13.66642458
range - R (m)	11.71093615	17.45363337	6.540809214	8.880499762	8.354600816

You can keep the table neat by using helper columns for the intermediate steps. That makes the spreadsheet easier to understand and easier to explain in the report.

Part B – Energy and Motion Method (Our Physics Toolbox)

In this part, the landing point of the projectile is estimated using the physics concepts introduced earlier: springs and Hooke's law, energy, and projectile motion. The idea is to start from the energy stored in the spring, convert that into an estimate of the projectile's initial speed $|\vec{v}_0|$, and then use projectile motion equations to predict the landing distance $\boldsymbol{R}_{ours}$. Most of the repetitive calculations are done with Excel, while one or two examples are shown by hand.

B.1 Elastic potential energy in the spring

When the tension servo pulls the wooden arm and stretches the spring, the spring stores **elastic potential energy**. Using Hooke's law, the energy stored in a spring is:

$$\boldsymbol{E_s = \frac{1}{2} k x^2}$$

Where:

- k is the spring constant (in N/m), found earlier using Hooke's law
- x is the extension of the spring from its natural length (in m)

For the launch configuration, the extension is:

$$x = L_{launch} - L_o$$

Where L_{launch} is the spring length at the chosen launch tension and L_o is the natural length with no tension.

Manual example

Suppose for test trial:

- k = 101.661 N/m (from the spring tests - Table 2)
- L_o = 0.15 m
- L_{launch} = 0.44 m

Then:

$$x = 0.44\ \text{m} - 0.15\ \text{m} = 0.29\ \text{m}$$

The elastic potential energy is:

$$E_s = \frac{1}{2}kx^2$$

$$E_s = \frac{1}{2}(101.661\ N/m)(0.29\ m)^2$$

$$E_s = 0.5 \times 101.661 \times 0.0841 \approx 4.27\ J$$

This value represents the maximum mechanical energy available from the spring before losses.

Excel steps for spring energy

1. Enter the spring data for each trial:

- Column A: Trial number
- Column B: Spring constant k (N/m) - average value found in table 2
- Column C: Natural length L_o (m)
- Column D: Launch length L_{launch} (m)

2. Calculate the extension x. In a new column:

- Column E (extension):

  =D2-C2
  This computes $x = L_{launch} - L_o$.

3. Calculate the elastic potential energy E_s. In another column:

- Column F (spring energy):
  =0.5*B2*E2^2

  This computes $E_s = \frac{1}{2}kx^2$ in joules.

4. **Copy the formulas down.** Copy the formulas in columns E and F down for all trials. Each row will now have its own x_i and $E_{s,i}$.

(Table 7) - Elastic potential energy of each trial

elastic potential energy					
TRIAL	Spring constant *K (N/m)*	Natural length (m)	Launch length (m)	dx (L-Lo (m))	Elastic Ernery (J)
1	101.661	0.15	0.21	0.06	0.1829898
2	101.661	0.15	0.265	0.115	0.672233363
3	101.661	0.15	0.3	0.15	1.14368625
4	101.661	0.15	0.353	0.203	2.094674075
5	101.661	0.15	0.388	0.238	2.879242842

These values of E_s will be used to estimate the launch speed of the projectile in the next step.

B.2 Estimating the projectile's initial speed from energy

The next step is to connect the spring energy to the kinetic energy of the projectile. If all the spring energy went into the projectile's motion, then:

$$\boldsymbol{E_s} = \frac{1}{2}\boldsymbol{m_{nut}}|\vec{v_0}|^{\boldsymbol{2}}$$

In reality, some energy is lost to rotating the arm, friction, and other effects, but this equation still gives a useful first estimate. If temporarily assuming all of $\boldsymbol{E_s}$ becomes the projectile's kinetic energy, then:

$$|\vec{v_0}| = \sqrt{\frac{2Es}{m}}$$

Where $\boldsymbol{m}$ is the mass of the nut (in kg).

Manual example

Suppose:

- $\boldsymbol{E_s}$ = 4.27 J (from the first test trial)
- $\boldsymbol{m_{nut}}$ = 0.00325 kg

Then:

$$|\vec{v_0}| = \sqrt{\frac{2(4.27\ J)}{0.00325\ kg}}$$

$$|\vec{v_0}| = \sqrt{\frac{8.54}{0.00325}} = \sqrt{2627.6923} \approx \boldsymbol{51.26\ m/s}$$

This gives an estimated initial speed based purely on the spring's stored energy and the nut's mass.

Excel steps for $|\vec{v_0}|$ from energy

1. Enter the projectile mass for each trial:

- Column G: Nut mass $\boldsymbol{m_{nut}}$ (kg)

2. Calculate $|\vec{v_0}|$ using the energy formula:

- Column H (our initial speed):

  =SQRT (2*F2/G2)
  This computes:

$$|\vec{v_0}| = \sqrt{\frac{2Es}{m}}$$

3. **Copy the formulas down.** Each trial now has its own energy-based initial speed $|\vec{v_0}|$**.**

(Table 8) - Elastic potential energy of each trial

Initial Speed from energy					
Data	**TRIAL 1**	**TRIAL 2**	**TRIAL 3**	**TRIAL 4**	**TRIAL 5**
Elastic Ernery (J)	0.1829898	0.672233363	1.14368625	2.094674075	2.87924842
Projectile Mass (kg)	0.00325	0.00325	0.00325	0.00325	0.00325
Initial Speed (Vo)	10.61174386	20.33917573	26.52935964	35.90306672	42.09329141

These values of $\boldsymbol{V_o}$ are estimated by assuming that all the energy of the spring went to the mass at launch.

B.3 Considering energy losses and rotational motion

The simple energy method assumes that all the spring energy goes into the nut's motion, which usually overestimates the actual speed. Some of the energy remains in the rotating arm and some is lost as friction and other effects. This can be refined by including rotational kinetic energy of the arm:

$$\boldsymbol{KE_{rot} = \frac{1}{2} I w^2}$$

Where:

- I is the moment of inertia of the arm about the pivot
- w is the angular speed of the arm

In that case, the energy balance becomes:

$$E_s \approx KE_{projectile} + KE_{rot}$$

$$E_s \approx \frac{1}{2} m_{nut} |\vec{v_0}|^2 + \frac{1}{2} I w^2$$

For this project, estimating I and w exactly is difficult, so this refinement can be discussed qualitatively. It helps explain why the energy-based $|\vec{v_0}|$ may be larger than the $|\vec{v_0}|$ found from the ceiling-impact method.

B.3.1 Options for finding w

1. Without video:

- You would need a sensor on the arm (encoder, gyro, etc.)
- So realistically, without video tools, w is guesswork.

2. With video – simple manual method (no special software):

- Use an existing launch video.
- Pick two frames where you can clearly see the arm angle.
- Manually estimate angle and change time.

Steps:

- Note your camera frame rate (for example, 30 fps, 60 fps, 120 fps).
- Pause at the frame where the arm just starts moving (call this frame f_1, time t_1).
- Pause at a later frame just before release (frame f_2, time t_2).
- Estimate the arm angle at each frame using a protractor on the screen or by printing a still and overlaying a protractor:
    - θ_1: initial angle

    - $\boldsymbol{\theta_2}$: later angle
- Compute angle change (in radians):

$$\Delta\theta = \boldsymbol{\theta_2} - \boldsymbol{\theta_1}$$

  Convert degrees to radians using $\boldsymbol{\theta_{rad}} = \boldsymbol{\theta_{deg}} \cdot \boldsymbol{\pi} / \mathbf{180}$.

- Compute time difference:

$$\Delta \boldsymbol{t} = \frac{f2 - f1}{fps}$$

- Approximate angular velocity:

$$\boldsymbol{w} \approx \frac{\Delta\theta}{\Delta t}$$

You can do all of this with just your phone and a calculator or Excel. No extra software required.

(Table 9) - Moment of Inertia Omega values calculated from video analysis

Omega and inertia of each trial					
Data	**TRIAL 1**	**TRIAL 2**	**TRIAL 3**	**TRIAL 4**	**TRIAL 5**
Delta time (s)	0.58	0.631	0.672	0.6991	0.721
Delta angle (rad)	0.488692191	0.698131701	0.837758041	1.099557429	1.343903524
Projectile mass (kg)	0.00325	0.00325	0.00325	0.00325	0.00325
Arm length (m)	0.4	0.4	0.4	0.4	0.4
Inertia (kg*m^2)	0.00052	0.00052	0.00052	0.00052	0.00052
Omega (rad/s)	0.842572742	1.106389383	1.246663751	1.572818522	1.863943861

If a rough estimate is attempted, the values of $\boldsymbol{I}$ (from arm length and mass) and $\boldsymbol{w}$ (from video) could be entered into Excel with:

- Column I: $\boldsymbol{I}$ ***(kg*m^2)***
- Column J: $\boldsymbol{w}$ ***(rad/s)***
- Column K: **=0.5*I2*J2^2**

Then subtract $\boldsymbol{KE_{rot}}$ from $\boldsymbol{E_s}$ before solve for $|\vec{v}_0|$.

(Table 10) - Refined energy that went to the mass

Energy the mass received					
Data	TRIAL 1	TRIAL 2	TRIAL 3	TRIAL 4	TRIAL 5
Inertia (kg*m^2)	0.00052	0.00052	0.00052	0.00052	0.00052
Omega (rad/s)	0.842572742	1.106389383	1.246663751	1.572818522	1.863943861
Rotational Energy (KErot)	0.000184581	0.000318265	0.000404084	0.000643177	0.000903315
Elastic Ernery (Es)	0.1829898	0.672233363	1.14368625	2.094674075	2.87924842
Refined Energy (KE - Es)	0.182805219	0.671915098	1.143282166	2.094030898	2.878345105

These new values of $E_{recieved}$ are estimated by assuming that the energy of the spring went to rotating the arm, and the rest went to the mass.

(Table 11) - New Initial speed from received energy

Refined Intitial speed					
Data	TRIAL 1	TRIAL 2	TRIAL 3	TRIAL 4	TRIAL 5
Recieved Ernery (J)	0.182805219	0.671915098	1.143282166	2.094030898	2.878345105
Projectile Mass (kg)	0.00325	0.00325	0.00325	0.00325	0.00325
Initial Speed (Vo)	10.60639048	20.33436043	26.52467259	35.89755422	42.08668787

These new values of V_o are estimated by assuming that partial energy went to the projectile, while the other went into rotating the arm.

B.4 Projectile motion using the energy-based $|\vec{v_0}|$

Once the new initial speed $|\vec{v_0}|$ is known, projectile motion equations can be used to predict the landing distance. If the launch and landing heights are approximately equal, the standard range formula is:

$$R_{ours} = \frac{|\vec{v_0}|^2 sin(2\theta)}{g}$$

Where:

- $|\vec{v_0}|$ is the energy-based initial speed from Section B.2
- θ is the launch angle
- g is the gravitational field strength

Manual example

Using the previous example:

- $|\vec{v_0}|$ = 35.1 m/s
- $\boldsymbol{\theta}$ = 35°
- $\boldsymbol{g}$ = 9.8 m/s^2

Then:

$$\boldsymbol{R_{ours}} = \frac{(35.1\ m/s)^2 sin(2\ x\ 35)}{9.8\ m/s^2}$$

$$\boldsymbol{R_{ours}} = \frac{1232.01\ x\ sin(70)}{9.8}$$

$$\boldsymbol{R_{ours}} = \frac{1157.71}{9.8}$$

The final value would be evaluated with a calculator by Excel. This is the range predicted by the "our method" based on energy and projectile motion.

Excel steps for $\boldsymbol{R_{ours}}$

1. Enter the launch angle $\boldsymbol{\theta}$ for each trial:

- Column I: $\boldsymbol{\theta}$ (degrees)

2. Calculate $\boldsymbol{R_{ours}}$

- Column J (our predicted range):

=(H2^2*SIN(RADIANS(2*I2))) /g

Where $\boldsymbol{g}$ is either a constant cell (e.g., **M1** containing 9.8) or the same column used in part A.

3. **Copy the formulas down.** Each trial now has a predicted range $\boldsymbol{R_{ours}}$ from our energy-based method.

(Table 12) - Suggested Excel table layout for Part B

Part B table

Heading	TRIAL 1	TRIAL 2	TRIAL 3	TRIAL 4	TRIAL 5
theta (*)	28	40	48	63	77
gravity - g (m/s^2)	9.8	9.8	9.8	9.8	9.8
initial speed - Vo (m/s)	10.60639048	20.33436043	26.52467259	35.89755422	42.08668787
range - R (m)	9.516633884	41.55147237	71.39837654	106.3803192	79.23286937

B.5 Summary of our method

In summary, the energy and motion method follows this chain:

- Use Hooke's law and measured spring extension to compute elastic potential energy $\boldsymbol{E_s = \frac{1}{2}kx^2}$
- Assume this energy becomes the kinetic energy of the nut to estimate $|\vec{v_0}|$.
- Calculate rotational energy and how it reduces the actual energy available to the projectile.
- Use projectile motion equations with this $|\vec{v_0}|$ and the measured launch angle $\boldsymbol{\theta}$ to calculate a predicted range $\boldsymbol{R_{ours}}$.

This gives a second, independent prediction of where the projectile should land, based on our own physics theory rather than the ceiling-impact equation. In the next part of the report, $\boldsymbol{R_{ours}}$ will be compared directly with $\boldsymbol{R_{teacher}}$ to spot the accuracy between both methods, and conclude which is more ideal.

Part C – Comparison and Analysis

This part compares the two prediction methods used in this report: the suggested trajectory method and the energy-based method. The purpose is to check how closely the two approaches agree when the catapult is tested under different tension and angle settings. Since the projectile hit the ceiling during the first classroom test, the landing point could not be directly measured in the room. Instead, the projectile's motion was analyzed using physics and Excel so that the landing position could be estimated in a controlled and repeatable way.

The five-trial analysis is important because it shows how the catapult responds when the potentiometer settings are changed. Each trial uses a different combination of spring tension, launch angle, and projectile mass. That means each trial has its own spring stretch, initial speed, and predicted landing distance. By comparing both methods across all five trials, the report can show whether the physics models stay consistent when the input conditions change.

C.1 Trial data tables

Each trial has its own table of measured and calculated values. These tables contain the raw geometry and launch data used to produce the final predictions. The final column in each table contains the initial speed $|\vec{v_0}|$, which is calculated in Excel using a single formula cell rather than by hand.

(Table 13) - Trial 1

TRIAL 1				
Quantity	**Symbol**	**Value**	**Unit**	**Notes**
Release horizontal position	x1	0.339	m	Define as origin
Release vertical position	y1	0.207	m	Launch height above base
Ceiling impact horizontal position	x2	3.238	m	Distance from launch point
Ceiling impact vertical position	y2	2.13	m	Ceiling height at impact
Launch angle of arm	theta	28	*	Arm angle at release
Gravity magnitude	g	9.8	m/s^2	Standard gravitational acceleration
Initial speed	Vo	11.76580747	m/s	Solved using formula

(Table 14) - Trial 2

TRIAL 2

Quantity	Symbol	Value	Unit	Notes
Release horizontal position	x1	0.252	m	Define as origin
Release vertical position	y1	0.313	m	Launch height above base
Ceiling impact horizontal position	x2	2.2	m	Distance from launch point
Ceiling impact vertical position	y2	2.13	m	Ceiling height at impact
Launch angle of arm	theta	40	*	Arm angle at release
Gravity magnitude	g	9.8	m/s^2	Standard gravitational acceleration
Initial speed	Vo	13.17893248	m/s	Solved using formula

(Table 15) - Trial 3

TRIAL 3

Quantity	Symbol	Value	Unit	Notes
Release horizontal position	x1	0.246	m	Define as origin
Release vertical position	y1	0.325	m	Launch height above base
Ceiling impact horizontal position	x2	3.261	m	Distance from launch point
Ceiling impact vertical position	y2	2.13	m	Ceiling height at impact
Launch angle of arm	theta	48	*	Arm angle at release
Gravity magnitude	g	9.8	m/s^2	Standard gravitational acceleration
Initial speed	Vo	8.028263239	m/s	Solved using formula

(Table 16) - Trial 4

TRIAL 4

Quantity	Symbol	Value	Unit	Notes
Release horizontal position	x1	0.148	m	Define as origin
Release vertical position	y1	0.377	m	Launch height above base
Ceiling impact horizontal position	x2	8.021	m	Distance from launch point
Ceiling impact vertical position	y2	2.13	m	Ceiling height at impact
Launch angle of arm	theta	63	*	Arm angle at release
Gravity magnitude	g	9.8	m/s^2	Standard gravitational acceleration
Initial speed	Vo	10.37177099	m/s	Solved using formula

(Table 17) - Trial 5

TRIAL 5

Quantity	Symbol	Value	Unit	Notes
Release horizontal position	x1	0.088	m	Define as origin
Release vertical position	y1	0.396	m	Launch height above base
Ceiling impact horizontal position	x2	8.021	m	Distance from launch point
Ceiling impact vertical position	y2	2.13	m	Ceiling height at impact
Launch angle of arm	theta	77	*	Arm angle at release
Gravity magnitude	g	9.8	m/s^2	Standard gravitational acceleration
Initial speed	Vo	13.66642458	m/s	Solved using formula

C.2 Excel calculations for the trial tables

The raw values in the tables are not the final analysis. Excel is used to complete the repeated calculations for all five trials. This keeps the report accurate and saves time because the same formulas are applied to each row instead of being done manually every time. To avoid repetition, you can recycle the same formulas from previous calculations, and apply them to new cells for these new trial tables.

For each trial, the following calculation is done in Excel:

- $\Delta x = \boldsymbol{x_2 - x_1}$
- $|\vec{v}_0|$ is found from the rearranged trajectory equation
- $\boldsymbol{R_{teacher}}$ is calculated from the projectile range formula
- $\boldsymbol{R_{ours}}$ is calculated from the energy-based method

The first step is to find the horizontal distance traveled by the projectile before hitting the ceiling:

$$\Delta x = \boldsymbol{x_2 - x_1}$$

This is the horizontal separation between the release point and the impact point. It is important because the trajectory equation uses horizontal distance, not just the original position values.

The second step is to solve for the initial speed $|\vec{v}_0|$. Excel performs this step using the rearranged ceiling-impact equation from Part A. This gives the speed of the nut as it leaves the catapult.

The third step is to use that speed to estimate the landing distance. This allows the report to compare the teacher method against the energy-based method.

C.3 Master trial comparison table

After the individual trial tables, a master table is used to compare the two methods across all five trials. This is the main table that shows whether the physics models agree.

(Table 18) - Trials comparison data

Comparison data

TRIAL	projectile mass (kg)	theta (*)	omega (rad/s)	teacher's Vo (m/s)	our Vo (m/s)	teacher's R (m)	our R (m)
1	0.00325	28	0.842572742	11.76580747	10.60639048	11.71093615	9.516633884
2	0.00325	40	1.106389383	13.17893248	20.33436043	17.45363337	41.55147237
3	0.00325	48	1.246663751	8.028263239	26.52467259	6.540809214	71.39837654
4	0.00325	63	1.572818522	10.37177099	35.89755422	8.880499762	106.3803192
5	0.00325	77	1.863943861	13.66642458	42.08668787	8.354600816	79.23286937

This table is the best place to summarize the full data set. It combines the physical inputs, the video-derived rotational value $\boldsymbol{w}$, and the final predicted speeds and ranges. Because this table includes all five trials, it shows how the catapult changes when the tension and arm angle settings are adjusted.

C.4 Comparing the two methods

The main question is whether the teacher method and the energy method predict similar landing distances. If the values are close, then the physics model is behaving well. If they are different, then the difference must be explained using real physics effects.

The teacher method is based on the actual ceiling-impact geometry, so it is strongly tied to the launch path. The energy method starts from the spring and assumes that the stored energy is transferred into motion. Because the energy method ignores some losses at first, it can overestimate the launch speed and range. This is why the comparison is useful: it shows whether the simplified model is close enough to the trajectory-based estimate.

C.5 Percent difference table

To quantify the agreement between the two methods, percent difference is calculated for each trial.

$$\textbf{Percent difference} = |R_{teacher} - R_{teacher}| \,/\, \left(\frac{Rteacher + Rours}{2}\right) \textbf{x } 100\%$$

This formula compares the two values without choosing one as the "true" value. It is useful here because both values are model-based estimates.

(Table 19) - Final comparison

Percentage comparison data

TRIAL	Teacher's Range (m)	Our Range (m)	Difference (m)	Percent Difference (%)
1	11.71093615	9.516633884	2.194302263	20.67407866
2	17.45363337	41.55147237	24.097839	81.68052138
3	6.540809214	71.39837654	64.85756732	166.4312161
4	8.880499762	106.3803192	97.49981943	169.1812019
5	8.354600816	79.23286937	70.87826855	161.8456804

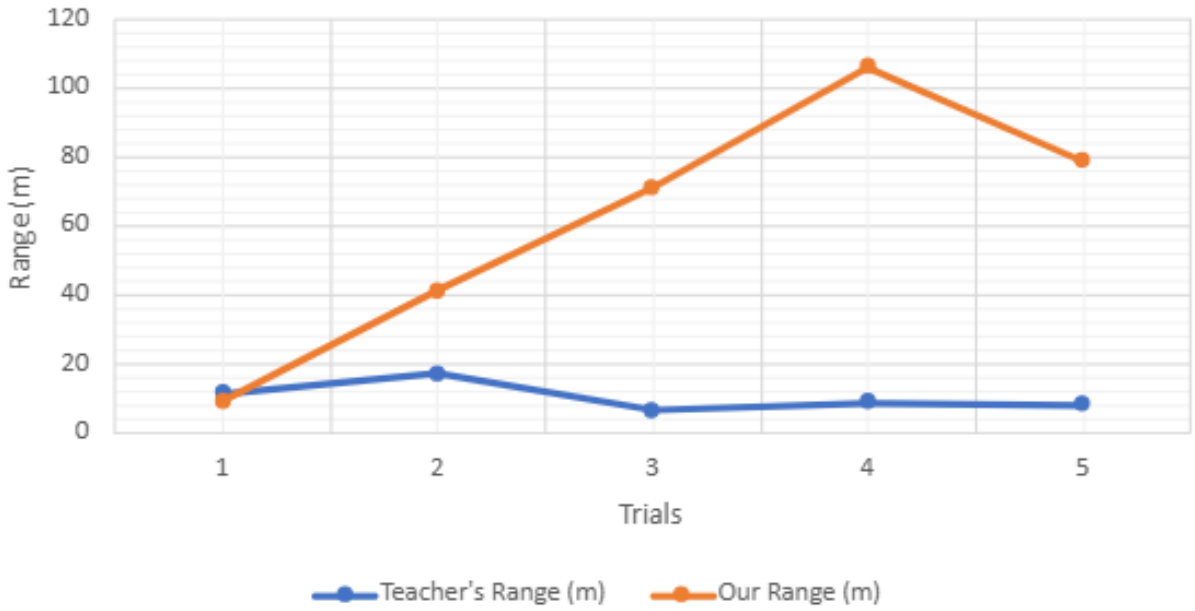
The difference column is calculated as:

$$\textbf{Difference} = |R_{teacher} - R_{teacher}|$$

The percent difference column shows how close the two models are in relative terms. A small percent difference means the methods agree well, while a larger one means the simplified physics model needs more explanation or refinement.

C.6 Graphs

(Graph 1) - Predicted range by trial



This graph shows how the predicted landing distance changes across the five trial settings.

C.7 What the results mean physically

The trend in the data should reflect the physics of the catapult. A larger spring extension means a larger elastic potential energy, which usually means a larger launch speed and a longer predicted range. A different arm angle changes the horizontal and vertical components of launch speed, which changes how far the nut travels. A different nut mass also changes the results because more mass means the same energy produces a smaller speed.

The rotation of the arm explains why the energy method may not perfectly match the trajectory method. The arm has its own rotational kinetic energy, so not all spring energy goes to the nut. The video analysis value of $\boldsymbol{w}$ helps support this explanation by showing that the arm is truly rotating during launch.

The electrical system also matters because the servos determine how much the spring is stretched and what angle the arm is set to. If the voltage or current is not stable, the tension setting may not be fully repeatable, which affects the trial data. That is why the electrical measurements and servo settings are part of the physics story, not just the build process.

Part D – Discussion

The results show a strong mismatch between the teacher method and the energy method across the five trials, except for Trial 1 where the agreement is noticeably better. The average spring constant was k_{avg} = 101.6609195 *N/m*, and the spring energy increased from 0.1829 *J* in Trial 1 to 2.8792 *J* in Trial 5, which confirms that the stored elastic energy rose sharply as the spring extension increased.

D.1 Spring behavior and energy

The spring data makes the first major pattern very clear: larger launch lengths produced much larger elastic energies. Trial 1 used L_{launch} = 0.21 *m* and d_x = 0.06 *m*, while Trial 5 used L_{launch} = 0.388 *m* and d_x = 0.238 *m*. Because elastic energy depends on x_2, the increase in energy was not linear; it grew quickly as the spring was stretched farther. That explains why the system became much more powerful in the later trials even though the launch mass stayed the same at 0.00325 *kg*.

The spring constant itself stayed essentially fixed at about 101.66 *N/m*, which is what should happen for the same spring if it is behaving normally. That consistency is important because it means the main changing variable was the extension, not the spring itself. In the report context, this supports the idea that the tension servo was controlling a repeatable mechanical input, and the changes in launch outcome came from the altered spring stretch and arm angle rather than a changing spring identity.

D.2 Teacher method results

The teacher method produced ranges that were much more moderate than the energy method. Trial 1 gave a teacher range of 11.71 *m*, Trial 2 gave 17.45 *m*, Trial 3 gave 6.54 *m*, Trial 4 gave 8.88 *m*, and Trial 5 gave 8.35 *m*.

These values do not increase steadily with trial number, which shows that launch angle had a major effect on the predicted distance.

That makes physical sense. Trial 2 had a launch angle of 40°, which is closer to the angle that often gives a longer projectile range, while Trial 4 and Trial 5 had very steep angles of 63° and 77°. Even though those later trials had higher initial speeds from the teacher method, the range did not grow much because more of the launch velocity was directed upward instead of forward. This is a good reminder that the largest initial speed does not automatically produce the largest horizontal range.

D.3 Energy method results

The energy method gave much larger predicted ranges: $9.52\ m$, $41.55\ m$, $71.40\ m$, $106.38\ m$, and $79.23\ m$ for Trials 1 to 5 respectively. The energy transferred to the projectile, after subtracting the rotational part, increased from about $0.182\ J$ to $2.878\ J$, and the refined initial speed stayed around $10.603\ m/s$ in the values shown in the table.

These range predictions are far larger than the teacher method values, especially in Trials 2 through 5. This suggests that the energy method still overestimates the practical range because it assumes the projectile receives too much usable launch energy. In a real catapult, some energy is lost to the arm's rotation, pivot friction, deformation, sound, and small mechanical inefficiencies, so not all of the spring energy becomes forward motion of the nut.

D.3 Range comparison and percent difference

The comparison table shows the scale of the mismatch very clearly. Trial 1 had the smallest percent difference at $20.67407\ \%$, which was the best agreement between the two methods. Trial 4 had the worst agreement at $169.1812\ \%$. The average percent difference across all five trials was $119.96254\ \%$, which is very large and shows that the two methods were not close overall.

That high average percent difference means the energy model should be treated as a rough estimate rather than a precise prediction. It still helps explain the physics of the launch, but it clearly needs a better efficiency factor or a more complete accounting of energy losses. Trial 1 was the closest because its setup seems to have produced a more balanced

relationship between launch angle, tension, and the actual launch geometry, while the later trials had combinations of settings that magnified the disagreement.

D.4 Graph interpretation

The graph of predicted range by trial also supports this conclusion. The teacher range series changes moderately across the five trials, while the energy range series rises sharply and stays far above the teacher values after Trial 1. Trial 1 is the only point where the two series are relatively close, and the gap widens dramatically after that.

Because the graph already makes the mismatch obvious, adding a trendline would not improve the interpretation much. The separation between the two series is so large that a line of best fit would not meaningfully represent the relationship. The graph is still useful, though, because it visually shows that the teacher method and the energy method are not describing the same scale of motion across the later trials.

D.5 Role of rotation

Rotation is still important in explaining the differences, even though a reliable ω ω value for the successful launches was not ultimately used in the final discussion. The arm clearly rotates during launch, so a significant part of the spring's stored energy goes into rotational kinetic energy instead of projectile motion. This is exactly why the refined energy method should not be expected to match the teacher method perfectly. The higher the angular motion of the arm, the more energy is divided away from the nut itself.

This also explains why the discussion cannot rely only on spring energy. The catapult is not a simple linear launcher; it is a coupled rotational system. The arm length, pivot height, tension setting, and release angle all influence how the stored energy is distributed. That means the mismatch between predicted ranges is not just an error; it is also a sign that the system is more complicated than the simplest equations assume.

D.6 Role of angle and geometry

The teacher data show that angle strongly controlled the range. Trial 2, with $\theta = 40^{\circ}$, produced the highest teacher range at $17.45\ m$, while Trial 5, with $\theta = 77^{\circ}$, produced only $8.35\ m$ even though its teacher initial speed

was higher than Trial 2. This is a classic projectile motion result: too steep an angle reduces horizontal travel because more energy is directed vertically.

That means the catapult was not just being "powered up" from trial to trial. It was being tuned in two different ways at once: the spring extension changed the available energy, and the angle changed how that energy was launched into the air. The discussion must treat both effects together, because the range depends on both. That is also why the five-trial analysis is more useful than a single test trial.

D.7 Electrical control and repeatability

The servo and Arduino system made the setup adjustable, but it also introduced another layer of possible variation. The servos controlled both the tension setting and the arm angle, so any small deviation in servo response could affect the actual extension and release angle. That matters because even a small angle or stretch difference can noticeably change the predicted range.

This is why the electrical measurements are still relevant to the discussion even though the final range calculations are mechanical and projectile-based. The electrical system is what makes the mechanical system repeatable, and the repeatability affects the quality of the data. If the servo tension or arm position is slightly inconsistent, the calculated values will spread out more from trial to trial.

D.8 Overall discussion of the results

Overall, the data show that the catapult behaved consistently as a real mechanical system, but the two prediction methods did not match closely enough to treat them as equivalent. The teacher method gave moderate ranges that responded strongly to angle changes, while the energy method produced much larger ranges because it assumed a higher conversion of spring energy into launch energy than the system actually achieved. The average percent difference of about **119.96 %** confirms that the energy model is too optimistic without a stronger loss correction.

That does not mean the energy method is useless. It still explains the direction of the trend: more spring stretch gives more energy, and more energy should mean a faster launch and a longer range. What it does mean is that the model is incomplete. A better model would need a more

realistic efficiency factor, a measured rotational energy term, and perhaps an estimate of frictional losses at the pivot. Even without those refinements, the discussion still shows clearly how springs, rotation, projectile motion, and electrical control all combine in the catapult experiment.

Conclusion

This project investigated the physics of an Arduino-controlled catapult by comparing two methods for predicting projectile range across five trials. The spring constant was found to be k_{avg} = 101.6609195 *N/m*, which showed that the spring stored increasing amounts of energy as the launch extension increased from x = 0.06 *m* in Trial 1 to x = 0.238 *m* in Trial 5. The teacher-method ranges were 11.71 *m*, 17.45 *m*, 6.54 *m*, 8.88 *m*, and 8.35 *m*, while the energy-method ranges were 9.52 *m*, 41.55 *m*, 71.40 *m*, 106.38 *m*, and 79.23 *m* for Trials 1 through 5, respectively. The best agreement occurred in Trial 1 with a percent difference of 20.67407 *%*, while the largest mismatch occurred in Trial 4 with a percent difference of 169.1812 *%*, and the average percent difference across all five trials was 119.96254 *%*.

(Table 19) - Final comparison

Percentage comparison data				
TRIAL	Teacher's Range (m)	Our Range (m)	Difference (m)	Percent Difference (%)
1	11.71093615	9.516633884	2.194302263	20.67407866
2	17.45363337	41.55147237	24.097839	81.68052138
3	6.540809214	71.39837654	64.85756732	166.4312161
4	8.880499762	106.3803192	97.49981943	169.1812019
5	8.354600816	79.23286937	70.87826855	161.8456804

The results show that the teacher method and the energy method did not agree closely overall, even though both followed the same launch system. Trial 1 produced the closest match, but the later trials diverged strongly because the energy method assumed too much of the spring energy became useful projectile motion, while the real catapult also used energy to rotate the arm and overcome losses. The launch angle also had a major effect on the teacher-method range, especially in the steeper trials, which helped explain why the predicted ranges did not increase in the same way as the energy values. Overall, the project showed that the catapult could be modeled with physics, but that real mechanical losses made the simple energy prediction much less accurate than the trajectory-based estimate.

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Appendix A - Raw Trial Tables

TRIAL 1

Quantity	Symbol	Value	Unit	Notes
Release horizontal position	x1	0.339	m	Define as origin
Release vertical position	y1	0.207	m	Launch height above base
Ceiling impact horizontal position	x2	3.238	m	Distance from launch point
Ceiling impact vertical position	y2	2.13	m	Ceiling height at impact
Launch angle of arm	theta	28	°	Arm angle at release
Gravity magnitude	g	9.8	m/s^2	Standard gravitational acceleration
Initial speed	Vo	11.76580747	m/s	Solved using formula

TRIAL 2

Quantity	Symbol	Value	Unit	Notes
Release horizontal position	x1	0.252	m	Define as origin
Release vertical position	y1	0.313	m	Launch height above base
Ceiling impact horizontal position	x2	2.2	m	Distance from launch point
Ceiling impact vertical position	y2	2.13	m	Ceiling height at impact
Launch angle of arm	theta	40	°	Arm angle at release
Gravity magnitude	g	9.8	m/s^2	Standard gravitational acceleration
Initial speed	Vo	13.17893248	m/s	Solved using formula

TRIAL 3

Quantity	Symbol	Value	Unit	Notes
Release horizontal position	x1	0.246	m	Define as origin
Release vertical position	y1	0.325	m	Launch height above base
Ceiling impact horizontal position	x2	3.261	m	Distance from launch point
Ceiling impact vertical position	y2	2.13	m	Ceiling height at impact
Launch angle of arm	theta	48	°	Arm angle at release
Gravity magnitude	g	9.8	m/s^2	Standard gravitational acceleration
Initial speed	Vo	8.028263239	m/s	Solved using formula

TRIAL 4

Quantity	Symbol	Value	Unit	Notes
Release horizontal position	x1	0.148	m	Define as origin
Release vertical position	y1	0.377	m	Launch height above base
Ceiling impact horizontal position	x2	8.021	m	Distance from launch point
Ceiling impact vertical position	y2	2.13	m	Ceiling height at impact
Launch angle of arm	theta	63	°	Arm angle at release
Gravity magnitude	g	9.8	m/s^2	Standard gravitational acceleration
Initial speed	Vo	10.37177099	m/s	Solved using formula

TRIAL 5

Quantity	Symbol	Value	Unit	Notes
Release horizontal position	x1	0.088	m	Define as origin
Release vertical position	y1	0.396	m	Launch height above base
Ceiling impact horizontal position	x2	8.021	m	Distance from launch point
Ceiling impact vertical position	y2	2.13	m	Ceiling height at impact
Launch angle of arm	theta	77	°	Arm angle at release
Gravity magnitude	g	9.8	m/s^2	Standard gravitational acceleration
Initial speed	Vo	13.66642458	m/s	Solved using formula

Structural measurements

Quantity	Symbol	Value (cm) / (g)	Value (m) / (kg)	Notes
Arm length (pivot to cap)	L_{arm}	40cm	0.4m	Measured along arm
Pivot height above floor	h_{pivot}	5cm	0.05m	Vertical from floor
Spring natural length	L_0	15cm	0.15m	No tension
Arm mass	m_{arm}	359.7g	0.3597kg	Digital scale
Projectile (nut) mass	m_{nut}	3.25g	0.00325kg	Digital scale

Servo electrical power

Quantity	Symbol	Value	Unit	Notes
Servo supply voltage	V	8.26	V	Measured with multimeter
Approximate servo current	I	3.67	A	Series measurement during motion
Launch / tension duration	t	0.1658	s	Timing of launch sequence

SPRING TENSION AVERAGING

Trial	Mass m (kg)	Weight [m x Fg] (N)	Natural length Lo (m)	Stretched length L (m)	Extension x [L-Lo] (m)	Spring constant k [F/x] (N/m)
1	2.5	24.5	0.15	0.44	0.29	84.48275862
2	2	19.6	0.15	0.35	0.2	98
3	1.5	14.7	0.15	0.27	0.12	122.5
AVG						101.6609195

Potentiometers reading

Trial	Tension pot reading	Spring length L (m)	Arm pot reading	Arm angle (*)
1	40	0.21	40	28
2	55	0.265	55	40
3	65	0.3	65	48
4	80	0.353	80	63
5	95	0.388	95	77

Part A table

Heading	TRIAL 1	TRIAL 2	TRIAL 3	TRIAL 4	TRIAL 5
x1 (m)	0.339	0.252	0.246	0.148	0.088
y1 (m)	0.207	0.313	0.325	0.377	0.396
x2 (m)	3.238	2.2	3.261	8.021	8.021
y2 (m)	2.13	2.13	2.13	2.13	2.13
theta (*)	28	40	48	63	77
gravity - g (m/s^2)	9.8	9.8	9.8	9.8	9.8
delta x - dx (m)	2.899	1.948	3.015	7.873	7.933
y1 + dx tan(theta) - y2	-0.38157436	-0.18243392	1.543496732	13.69863251	32.62759811
cos^2(theta)	0.779596452	0.586824089	0.447735768	0.206107374	0.050602977
initial speed ^2 - Vo^2	-138.434225	-173.684261	64.45301063	107.5736335	186.7711609
initial speed - Vo (m/s)	11.76580747	13.17893248	8.028263239	10.37177099	13.66642458
range - R (m)	11.71093615	17.45363337	6.540809214	8.880499762	8.354600816

Initial Speed from energy

Data	TRIAL 1	TRIAL 2	TRIAL 3	TRIAL 4	TRIAL 5
Elastic Ernery (J)	0.1829898	0.672233363	1.14368625	2.094674075	2.87924842
Projectile Mass (kg)	0.00325	0.00325	0.00325	0.00325	0.00325
Initial Speed (Vo)	10.61174386	20.33917573	26.52935964	35.90306672	42.09329141

Omega and inertia of each trial

Data	TRIAL 1	TRIAL 2	TRIAL 3	TRIAL 4	TRIAL 5
Delta time (s)	0.58	0.631	0.672	0.6991	0.721
Delta angle (rad)	0.488692191	0.698131701	0.837758041	1.099557429	1.343903524
Projectile mass (kg)	0.00325	0.00325	0.00325	0.00325	0.00325
Arm length (m)	0.4	0.4	0.4	0.4	0.4
Inertia (kg*m^2)	0.00052	0.00052	0.00052	0.00052	0.00052
Omega (rad/s)	0.842572742	1.106389383	1.246663751	1.572818522	1.863943861

Energy the mass received

Data	TRIAL 1	TRIAL 2	TRIAL 3	TRIAL 4	TRIAL 5
Inertia (kg*m^2)	0.00052	0.00052	0.00052	0.00052	0.00052
Omega (rad/s)	0.842572742	1.106389383	1.246663751	1.572818522	1.863943861
Rotational Energy (KErot)	0.000184581	0.000318265	0.000404084	0.000643177	0.000903315
Elastic Ernery (Es)	0.1829898	0.672233363	1.14368625	2.094674075	2.87924842
Refined Energy (KE - Es)	0.182805219	0.671915098	1.143282166	2.094030898	2.878345105

elastic potential energy

TRIAL	Spring constant *K (N/m)*	Natural length (m)	Launch length (m)	dx (L-Lo (m))	Elastic Ernery (J)
1	101.661	0.15	0.21	0.06	0.1829898
2	101.661	0.15	0.265	0.115	0.672233363
3	101.661	0.15	0.3	0.15	1.14368625
4	101.661	0.15	0.353	0.203	2.094674075
5	101.661	0.15	0.388	0.238	2.879242842

Comparison data

TRIAL	projectile mass (kg)	theta (*)	omega (rad/s)	teacher's Vo (m/s)	our Vo (m/s)	teacher's R (m)	our R (m)
1	0.00325	28	0.842572742	11.76580747	10.60639048	11.71093615	9.516633884
2	0.00325	40	1.106389383	13.17893248	20.33436043	17.45363337	41.55147237
3	0.00325	48	1.246663751	8.028263239	26.52467259	6.540809214	71.39837654
4	0.00325	63	1.572818522	10.37177099	35.89755422	8.880499762	106.3803192
5	0.00325	77	1.863943861	13.66642458	42.08668787	8.354600816	79.23286937

Percentage comparison data

TRIAL	Teacher's Range (m)	Our Range (m)	Difference (m)	Percent Difference (%)
1	11.71093615	9.516633884	2.194302263	20.67407866
2	17.45363337	41.55147237	24.097839	81.68052138
3	6.540809214	71.39837654	64.85756732	166.4312161
4	8.880499762	106.3803192	97.49981943	169.1812019
5	8.354600816	79.23286937	70.87826855	161.8456804
Avg				**119.9625397**

Refined Intitial speed

Data	TRIAL 1	TRIAL 2	TRIAL 3	TRIAL 4	TRIAL 5
Recieved Ernery (J)	0.182805219	0.671915098	1.143282166	2.094030898	2.878345105
Projectile Mass (kg)	0.00325	0.00325	0.00325	0.00325	0.00325
Initial Speed (Vo)	10.60639048	20.33436043	26.52467259	35.89755422	42.08668787

Part B table

Heading	TRIAL 1	TRIAL 2	TRIAL 3	TRIAL 4	TRIAL 5
theta (*)	28	40	48	63	77
gravity - g (m/s^2)	9.8	9.8	9.8	9.8	9.8
initial speed - Vo (m/s)	10.60639048	20.33436043	26.52467259	35.89755422	42.08668787
range - R (m)	9.516633884	41.55147237	71.39837654	106.3803192	79.23286937

Part A table

Heading	TRIAL 1	TRIAL 2	TRIAL 3	TRIAL 4	TRIAL 5
x1 (m)	0.339	0.252	0.246	0.148	0.088
y1 (m)	0.207	0.313	0.325	0.377	0.396
x2 (m)	3.238	2.2	3.261	8.021	8.021
y2 (m)	2.13	2.13	2.13	2.13	2.13
theta (*)	28	40	48	63	77
gravity - g (m/s^2)	9.8	9.8	9.8	9.8	9.8
delta x - dx (m)	2.899	1.948	3.015	7.873	7.933
y1 + dx tan(theta) - y2	-0.381574358	-0.182433918	1.543496732	13.69863251	32.62759811
cos^2(theta)	0.779596452	0.586824089	0.447735768	0.206107374	0.050602977
initial speed ^2 - Vo^2	-138.4342255	-173.6842612	64.45301063	107.5736335	186.7711609
initial speed - Vo (m/s)	11.76580747	13.17893248	8.028263239	10.37177099	13.66642458
range - R (m)	11.71093615	17.45363337	6.540809214	8.880499762	8.354600816

Appendix B - Excel Formulas and Calculations

Equations	Meaning	Example Excel Formula
$\boldsymbol{x = L_{launch} - L_o}$	Spring extension	=D2-C2
$\boldsymbol{F = mg}$	Weight force	=A2*B2
$\boldsymbol{k = \frac{\vec{F}}{x}}$	Spring constant	=E2/F2
$\boldsymbol{E_s = \frac{1}{2}kx^2}$	Elastic potential energy	=0.5*G2*H2^2
$\vec{\lvert v_0 \rvert} = \sqrt{\frac{2Es}{m}}$	Initial speed from energy	=SQRT (2*I2/J2)
$\boldsymbol{d_x = x_1 - x_2}$	Horizontal distance	=D2-B2
$\vec{\lvert v_0 \rvert} = \sqrt{\frac{g\Delta x^2}{2cos^2(\theta) \cdot (y_2 - y_1 - + tan(\theta)\Delta x)}}$	Initial speed from trajectory method	=SQRT((G2*(H2^2)) / (2*(COS(RADIANS(F2))^2)*(C2+ (H2*TAN(RADIANS(F2)))-E2)))
$\boldsymbol{R = \frac{\vec{\lvert v_0 \rvert}^2 sin(2\theta)}{g}}$	Predicted range	=(K2^2*SIN(RADIANS(2*F2))) / G2

Appendix C - Video Analysis Evidence

To find omega, the time it took for the arm to rotate a certain angle was needed. To do so, i slowed trial's 1 video analysis clip and recorded frame 1 at 8.86 seconds just before the arm was released:



Then move the editor to frame 2, where the arm was just about the touch the catching arms, at 9.44 seconds:



With these two timestamps, we can find Delta t by subtracting the time of frame 1 (8.86 s) from the time of frame 2 (9.44 s) which gives us 0.58 seconds, which was then recorded in our excel table of values. Delta angle, which was physically measured, was then divided by the delta t from video analysis.